

Joint and Cross-Modal Video-Audio Generation and Editing: A Unified Formulation and Design Taxonomy

Abhinav Sharma¹, Sai Karthik Navuluru², Wang Wei³, Daksh Dangi¹, Xiangbo Gao⁴,
Li Li⁵, Bo Ni⁶, Vardhan Dongre⁷, Junda Wu⁸, Xiyang Hu⁹, Jiuxiang Gu⁸,
Seunghyun Yoon⁸, Tong Yu⁸, Chien Van Nguyen¹⁰, Mohamed Elmoghany¹¹,
Nedim Lipka⁸, Hoda Eldardiry³, Hongjie Chen¹², Tyler Derr⁶, Thien Huu Nguyen¹⁰,
Zhengzhong Tu⁴, Nesreen K. Ahmed¹³, Franck Dernoncourt⁸, Ryan A. Rossi⁸

¹University of Massachusetts Amherst ²University of Texas at Dallas ³Virginia Tech
⁴Texas A&M University ⁵University of Southern California ⁶Vanderbilt University
⁷University of Illinois Urbana-Champaign ⁸Adobe Research ⁹Arizona State University
¹⁰University of Oregon ¹¹Stanford University ¹²Dolby Laboratories ¹³Cisco

Abstract

Video and audio are perceived together, yet most generative models treat them in isolation. We examine methods that model the two modalities jointly, generate one from the other, or edit them in a coupled manner, organized around a single question: how is the output kept coherent across modalities in time and semantics? A unified formulation casts joint generation, cross-modal generation, and joint editing as three problems defined on a single distribution over audio-visual pairs, and a taxonomy compares methods along five design axes. To our knowledge, this is the first overview to systematically taxonomize joint audio-visual *editing*, which we map as nine edit categories spanning 28 edit types. We describe methods, datasets, and metrics for each setting and close with the open problems we view as most consequential.

1 Introduction

Sound and picture are perceived as a single percept: a door closing and its impact sound coincide, and even a small offset reads as an error. Generative models of video and audio, however, developed largely in isolation, so two strong unimodal capabilities, combined naively, produce streams that do not agree. A growing body of work instead treats the pair as coupled, in three settings grouped by *output*: (i) both modalities generated together; (ii) one generated from the other, as in soundtracking a silent video; and (iii) an existing pair edited so a change in one modality propagates to the other. All three share one demand—coherence across modalities in time and semantics—which we take as our organizing principle.

Contributions. Existing overviews treat video generation, audio generation, or audio-visual understanding in isolation (Qin et al., 2026). This work covers generation and editing of the two streams as a coupled pair. Our contributions are summarized as follows:

- **First systematic taxonomy of joint audio-visual editing.** To our knowledge, this is the first overview to systematically taxonomize the editing of video and audio as a coupled pair, in which an edit specified in one modality must propagate to the other; we map this space as a taxonomy of nine categories and 28 edit types, each with representative operations and use cases (Table 4, Section 5).
- **A unified formulation and five-axis design taxonomy.** Section 2 casts joint generation, cross-modal generation, and joint editing as three problems over one distribution on audio-visual pairs, and Table 1 organizes the literature along five design axes.
- **Open problems grounded in the formulation.** Section 13 states the open problems—long-horizon coherence, fine-grained control, physical plausibility, and evaluation—as instances of one underlying challenge: raising cross-modal alignment while preserving per-stream quality.

Scope. A method is in scope when at least one of video or audio is among its outputs and the other appears in its pipeline; single-modality generation and audio-visual understanding without a generative or editing component are excluded. The closest overlapping overview, a concurrent review of audio-visual intelligence in foundation models (Qin et al.,

2026), treats neither editing nor our organizing device. Figure 1 maps the organization.

2 Problem Formulation and Notation

Notation. A video is $v \in \mathbb{R}^{T_v \times H \times W \times 3}$, where T_v is the number of frames and H, W are the spatial dimensions. An audio signal is $a \in \mathbb{R}^{T_a}$, where T_a is the number of audio samples. A condition is c , with subscripts for specific modalities: c_t for text, c_i for an image, c_s for speech, and c_m for music. A generative model is p_θ with parameters θ . Generated outputs are written $\hat{v}, \hat{a}$, and edited outputs v', a' . We write $v^{(t)}$ for the t -th frame and use ϕ_v and ϕ_a for encoders that map video and audio into latent or shared representation spaces, with $z_v = \phi_v(v)$ and $z_a = \phi_a(a)$. Table 2 collects the symbols used throughout this work.

An audio-visual clip is a pair $x = (v, a)$ with $v \in \mathbb{R}^{T_v \times H \times W \times 3}$ and $a \in \mathbb{R}^{T_a}$ sharing a time interval $[0, \tau]$, so frame and sample index refer to the same physical time; $\mathcal{X} = \mathcal{V} \times \mathcal{A}$ is the space of such pairs.

Definition 1 (Audio-visual correspondence). (v, a) corresponds when the streams agree *semantically*—the sources visible are the sources audible—and *temporally*—each acoustic event is localized to the frames of its visual cause: for an alignment score $\mathcal{S} : \mathcal{X} \rightarrow \mathbb{R}$, natural clips satisfy $\mathcal{S}(v, a) \geq \mathcal{S}(v, \tilde{a})$ for mismatched or time-shifted $\tilde{a}$.

Definition 1 is what separates the joint and cross-modal setting from two independent unimodal problems. A model that produces a high-quality $\hat{v}$ and a high-quality $\hat{a}$ but assigns them low $\mathcal{S}(\hat{v}, \hat{a})$ is perceived as broken, even when each stream is convincing on its own. The methods in Sections 5 through 7 differ primarily in how they raise $\mathcal{S}$ while keeping the per-modality quality high.

Definition 2 (Latent representations). Methods typically operate on $z_v = \phi_v(v)$ and $z_a = \phi_a(a)$ with decoders ψ_v, ψ_a ; p_θ is defined over (z_v, z_a) , and ϕ_v, ϕ_a fix the coupling space.

Problem 1 (Joint audio-visual generation). Given c (possibly $\emptyset$), learn $p_\theta(v, a | c)$ whose samples are faithful to c , of high per-modality quality, and of high $\mathcal{S}(\hat{v}, \hat{a})$.

The defining requirement in Problem 1 is that the parameterization of $p_\theta(v, a | c)$ encode the dependency between v and a . A model that factorizes as $p_\theta(v | c) p_\theta(a | c)$ with no further coupling treats the two modalities as conditionally independent

given c and cannot, in general, raise $\mathcal{S}$ beyond what c already determines. Joint methods therefore introduce coupling either in the architecture, through shared parameters or cross-modal attention, or in the objective, through a term that rewards correspondence.

Problem 2 (Cross-modal generation). Given one modality, generate the other so that the pair corresponds: video-to-audio learns $p_\theta(a | v)$, audio-to-video learns $p_\theta(v | a)$.

Problem 2 differs from Problem 1 in what is given. In the joint setting both modalities are outputs of a single distribution; in the cross-modal setting one of them is the condition. The contrast is concrete: a joint text-to-audio-visual model given $c_t = \text{“footsteps on gravel”}$ synthesizes both the visual scene and the footstep sounds, whereas a cross-modal video-to-audio model given a silent video v of a person walking on gravel synthesizes only $\hat{a}$, with the footstep sounds aligned to the frames in which the foot contacts the ground.

Problem 3 (Joint audio-visual editing). Given (v, a) and an instruction e (text, mask, style, or identity), sample $(v', a') \sim p_\theta(v', a' | v, a, e)$ such that the edit is applied, reflected in both modalities so $\mathcal{S}(v', a')$ stays high, and content outside the targeted region is preserved.

Problem 3 adds two constraints absent from generation: the propagation of an edit across modalities, and the preservation of untouched content. As an example, given a clip of a person speaking and the instruction $e = \text{“change the speaker’s voice to a child’s voice,”}$ the model must produce a' with the new voice and v' in which the lip motion is consistent with a' , while leaving the background and identity intact. Editing thus inherits the correspondence requirement of generation and adds a fidelity-to-input requirement on top of it. Figure 2 summarizes the three problems as mappings from inputs to outputs, Table 3 instantiates them task by task, and Table 2 collects the notation.

3 Background and Preliminaries

This section introduces the building blocks that joint and cross-modal methods inherit from the unimodal setting: the generative model families that instantiate p_θ , the representations that realize Definition 2 for each modality, and the conditioning mechanisms that inject c .

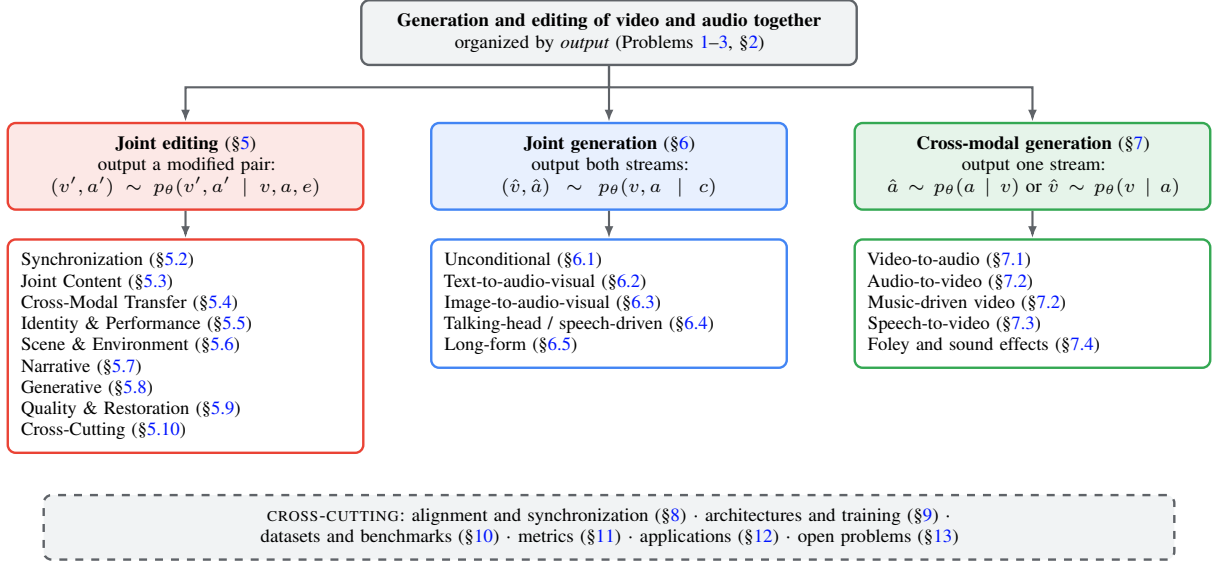


Figure 1: **Roadmap of this work.** Methods are grouped by the *output* they produce (Section 2): joint editing outputs a modified pair, joint generation outputs both streams, and cross-modal generation exactly one—the signals a method consumes vary freely within each family. The listed subsections cover each family; the gray strip collects the cross-cutting sections that apply to all three. Color marks the family (red editing, blue joint, green cross-modal) and is redundant with position and headings. Notation follows Table 2.

3.1 Generative Modeling Foundations

The methods we cover instantiate a generative model p_θ that approximates a target distribution over video, audio, or both. Four families recur: variational autoencoders (Kingma and Welling, 2014), generative adversarial networks (Goodfellow et al., 2014), autoregressive models (van den Oord et al., 2016), and diffusion and flow-matching models (Ho et al., 2020; Lipman et al., 2023). Each corresponds to a different parameterization of p_θ and a different training objective, and each has been adapted to the joint and cross-modal setting in its own way. Recent work has converged on diffusion and flow-matching backbones for both video and audio synthesis, and a large fraction of the methods in later sections build directly on top of one. We therefore use the denoising formulation as the running example: a forward process gradually corrupts a clean latent z_0 into noise, and the model learns to reverse it, optionally conditioned on c , by predicting the noise or the velocity at each step.

3.2 Video Representations

Several representational spaces realize ϕ_v in Definition 2, and the choice has direct consequences for the design of p_θ . Pixel-space models operate on v directly. Latent-space models, following the latent diffusion recipe (Rombach et al., 2022), first encode v into a lower-dimensional latent $z_v = \phi_v(v)$

and model the distribution over z_v rather than over raw pixels, using either a 2D autoencoder applied per frame with a separate temporal module, or a 3D autoencoder that compresses space and time jointly. Token-based representations take a further step and discretize v into a sequence of tokens, which is what enables autoregressive modeling. The three representations trade off fidelity, computational cost, and compatibility with the audio representation used alongside them, with the last factor mattering most in joint models that couple v and a in a shared space.

3.3 Audio Representations

An audio signal admits an analogous set of choices for ϕ_a . Waveform-domain models operate on a directly. Spectrogram-domain models first transform a into a time-frequency representation such as a mel spectrogram and model that image-like array, relying on a neural vocoder to recover the waveform (Kong et al., 2020). Latent audio codecs (Défossez et al., 2022) encode a into a sequence of discrete or continuous tokens and decode back to the waveform, playing a role analogous to latent video encoders. As with video, the representation chosen for audio interacts with the one chosen for video in joint models, since the two streams must be aligned in time and, for unified backbones, processed by a shared network. A recurring difficulty is the mis-

Table 1: **Complementary taxonomy of joint audio-video methods along five complementary design axes:** **GENERATION STRATEGY** (§4.1) captures how the two modalities are produced; **AUDIO REPRESENTATION** (§4.2) describes the latent space in which audio is generated; **VIDEO REPRESENTATION** (§4.3) describes the latent space in which video is generated; **ALIGNMENT ENFORCEMENT** (§4.4) identifies the stage at which audio-video alignment is imposed; and **PRETRAINING REUSE** (§4.5) characterizes the source of model weights. A check mark (✓) indicates that a method falls under the corresponding category within each axis. For the partially closed system Wan 2.5, the axes whose design is not publicly disclosed (alignment enforcement and pretraining reuse) are left without a mark.

	GENERATION STRATEGY (Section 4.1)					AUDIO REPRESENTATION (Section 4.2)				VIDEO REPRESENTATION (Section 4.3)				ALIGNMENT ENFORCEMENT (Section 4.4)					PRETRAINING REUSE (Section 4.5)		
	Single-Tower (§4.1.1)	Dual-Tower (§4.1.2)	Cascaded (§4.1.3)	Unified-Token (§4.1.4)	Guidance-Based (§4.1.5)	Waveform (§4.2.4)	Mel-Spectrogram (§4.2.3)	Continuous Latent (§4.2.1)	Discrete Tokens (§4.2.2)	Pixel (§4.3.4)	2D-VAE + Temporal (§4.3.2)	3D-VAE (§4.3.1)	Discrete Tokens (§4.3.3)	Cross-Attention (§4.4.1)	Shared Pos. Enc. (§4.4.2)	Discriminator (§4.4.3)	Classifier Guidance (§4.4.4)	Explicit Prior (§4.4.5)	From-Scratch (§4.5.1)	Single Pretrained (§4.5.2)	Dual Pretrained (§4.5.3)
JOINT AUDIO-VIDEO GENERATION																					
MM-Diffusion (Ruan et al., 2023)	×	✓	×	×	×	×	✓	×	×	✓	×	×	×	✓	×	×	×	×	✓	×	×
CoDi (Tang et al., 2023)	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	×	×	✓
Seeing-and-Hearing (Xing et al., 2024a)	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	✓	×	×	×	×	✓
AV-DiT (Wang et al., 2024b)	✓	×	×	×	×	×	×	✓	×	×	✓	×	×	×	✓	×	×	×	×	✓	×
MM-LDM (Sun et al., 2024)	✓	×	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	✓	×	×
Movie Gen (Polyak et al., 2024)	×	×	✓	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
SVG (Ishii et al., 2024)	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	×	×	✓
MMDiCo (Hayakawa et al., 2025)	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	✓	×	×	×	×	✓
SyncFlow (Liu et al., 2024b)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
JavisDiT (Liu et al., 2026a)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	×	×	×	×	✓	✓	×	×
JavisDiT++ (Liu et al., 2026b)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	×	✓	✓	×	×
BridgeDiT (Guan et al., 2025)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	×	✓
ALIVE (Guo et al., 2026b)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	×	×	×	✓	×
Ovi (Low et al., 2025)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
UniAVGen (Zhang et al., 2025)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
Animate-and-Sound (Wang et al., 2025c)	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	×	×	✓
CCL (Ma et al., 2026)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	×	✓
Hallo-Live (Li et al., 2026)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	×	✓
UniForm (Zhao et al., 2025)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	×	×	✓	×	×
Wan 2.5 (Alibaba, 2025)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×								
LTX-2 (HaCohen et al., 2026)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
MOVA (SII-OpenMOSS Team, 2026)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×
Apollo (Wang et al., 2026)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	×	×	✓	×	×
3MDiT (Li et al., 2025)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	×	×	×	✓	×
OmniForcing (Su et al., 2026)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	✓	×
JOINT AUDIO-VIDEO EDITING																					
Lang.-Guid. AV Edit (Liang et al., 2024b)	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	×	×	✓
AV-Edit (Guo et al., 2026a)	✓	×	×	×	×	×	×	✓	×	×	✓	×	×	✓	×	×	×	×	×	✓	×
JUST-DUB-IT (Chen et al., 2026)	×	✓	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	✓	×
EditYourself (Flynn et al., 2026)	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	✓	×	×	×	×	×	✓	×

Table 2: **Notation.** We summarize the symbols used throughout this work. Subscripts denote modality and primes denote edited outputs.

Symbol	Meaning
$v \in \mathbb{R}^{T_v \times H \times W \times 3}$	video, T_v frames of size $H \times W$
$a \in \mathbb{R}^{T_a}$	audio signal of T_a samples
$x = (v, a)$	audio-visual clip
$\mathcal{X} = \mathcal{V} \times \mathcal{A}$	space of audio-visual clips
fps, sr	frame rate, audio sampling rate
τ	shared clip duration
c	condition signal
c_t, c_i, c_s, c_m	text, image, speech, music
p_θ	generative model with parameters θ
$\theta_v, \theta_a, \theta_\times$	modality-specific and cross-modal parameters (§4.1)
Γ	sampling procedure (§4)
λ	guidance weight (§4.1.5)
$\hat{v}, \hat{a}$	generated video, audio
v', a'	edited video, audio
ϕ_v, ϕ_a	video, audio encoders
z_v, z_a	video, audio latents (Def. 2)
ψ_v, ψ_a	video, audio decoders
$\mathcal{S}(v, a)$	alignment score (Def. 1)
δ	synchronization tolerance (Def. 3)
e	edit instruction
$\mathcal{D}$	dataset (Def. 4)
$v^{(t)}$	t -th video frame

match in native rate between the two modalities, since audio is sampled far more densely in time than video is, and the latent rates must be reconciled for the streams to be coupled frame by frame.

3.4 Conditioning Mechanisms

Once the representations are fixed, a conditioning signal c enters the generative model through one of several mechanisms. Cross-attention injects c into intermediate features of the network, giving fine-grained, position-dependent control. Adaptive normalization modulates feature statistics based on c , providing a lighter and coarser form of control. Concatenation appends an encoded c to the input or to intermediate features. Classifier-free guidance (Ho and Salimans, 2022) steers samples toward c at inference time by mixing conditional and unconditional predictions. The form of c together with the mechanism through which it is injected determines how tightly the output follows the condition, and in the joint setting the same mechanisms are reused to let one modality condition the other.

3.5 Audio-Visual Correspondence in Practice

Definition 1 states correspondence as an abstract property; in practice it is operationalized through learned encoders that map a clip to a score $\mathcal{S}(v, a)$. Contrastive audio-visual encoders trained to pull matched pairs together and push mismatched pairs apart, of which ImageBind (Girdhar et al., 2023)

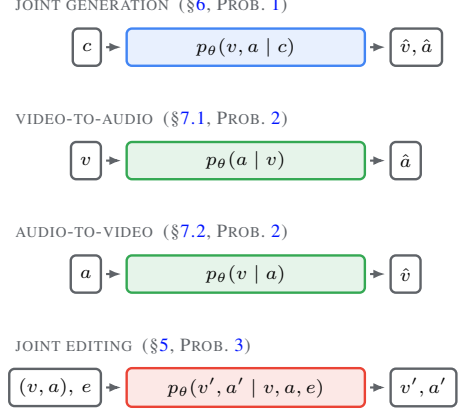


Figure 2: **The three problem settings as input-to-output mappings**, in the notation of Table 2: what varies across rows is only which signals are given (left) and which are generated (right). Joint generation outputs both streams from a condition c (which may be empty, text c_t , an image c_i , speech c_s , or music c_m); cross-modal generation outputs exactly one stream given the other; joint editing outputs a modified pair given a clip and an edit instruction e . Row color marks the family as in Figure 1. Every setting must satisfy the correspondence requirement of Definition 1.

is a widely reused instance, provide such a score—temporally focused variants additionally treat time-shifted pairs as negatives (Luo et al., 2023)—and several methods reuse these encoders either as a training signal or as a guidance term at inference. The remainder of this work is structured around how methods raise $\mathcal{S}$ while keeping per-modality quality high, since this is the property that distinguishes the joint and cross-modal problem from two unimodal ones.

4 A Five-Axis Design Taxonomy

Methods that solve Problems 1 through 3 differ along a small number of design axes that, taken together, account for most of the variation in the literature. We propose a taxonomy that categorizes methods along five such axes, summarized in Table 1: the generation strategy, the audio representation, the video representation, the alignment-enforcement mechanism, and the reuse of pre-trained weights.

Formally, a method is characterized by a tuple $(\phi, \theta, \mathcal{L}, \Gamma)$: the encoders $\phi = (\phi_v, \phi_a)$ of Definition 2, which fix the spaces $\mathcal{Z}_v, \mathcal{Z}_a$ in which generation occurs; the parameters θ of the model p_θ acting on those spaces; the training objective $\mathcal{L}$ used to train θ ; and the sampling procedure Γ that draws $(\hat{z}_v, \hat{z}_a)$ from p_θ . The five axes constrain different

Table 3: **A unified view of the tasks covered.** Each task is an instance of Problems 1 through 3: it models the joint distribution over audio-visual pairs or one of its conditionals. **Given** lists the observed signals, **Generated** the produced signals, and **Primary Correspondence** the form of audio-visual agreement (Def. 1) that dominates evaluation. The table doubles as a map from a task to the section that treats it. [†]For talking-head generation the model re-synthesizes the speech track: c_s conditions the output and $\hat{a}$ is emitted synchronized to $\hat{v}$, so speech appears as both condition and output.

Task	Given	Generated	Learned Distribution	Primary Correspondence	Section
Unconditional joint	$\emptyset$	v, a	$p_\theta(v, a)$	semantic + temporal	§6.1
Text-to-audio-video	c_t	v, a	$p_\theta(v, a \mid c_t)$	semantic + temporal	§6.2
Image-to-audio-video	c_i	v, a	$p_\theta(v, a \mid c_i)$	semantic + temporal	§6.3
Talking-head / speech-driven	c_i, c_s	$v, a^\dagger$	$p_\theta(v, a \mid c_i, c_s)$	lip-sync	§6.4
Music-driven video	c_m	v	$p_\theta(v \mid c_m)$	beat / rhythm	§7.2
Video-to-audio / Foley	v	a	$p_\theta(a \mid v)$	temporal (onset)	§7.1
Audio-to-video	a	v	$p_\theta(v \mid a)$	temporal	§7.2
Speech-to-video	c_s	v	$p_\theta(v \mid c_s)$	lip-sync	§7.3
Joint editing	v, a, e	v', a'	$p_\theta(v', a' \mid v, a, e)$	preserve + propagate	§5
Dubbing / re-voicing	v, a, e	v', a'	$p_\theta(v', a' \mid v, a, e)$	lip-sync	§5.2

components of this tuple. The generation strategy (§4.1) constrains how θ decomposes across the two streams; the audio and video representations (§4.2, §4.3) constrain the codomains of ϕ_a and ϕ_v ; the alignment-enforcement mechanism (§4.4) constrains the stage at which correspondence pressure is applied, which may be in θ , in the training objective $\mathcal{L}$, or in Γ ; and pretraining reuse (§4.5) constrains the initialization of θ . Because the axes constrain different components, a method makes a choice on each, and two choices on different axes are not alternatives to one another.

The axes are complementary rather than orthogonal. Each constrains a different component, so a method makes a choice along every axis, but the choices are not fully independent: a guidance-based strategy requires two frozen unimodal models and therefore entails dual pretraining, and pretraining reuse interacts with the representation axes in turn—every method covered here that reuses one or two pretrained backbones inherits its representation from them, and the backbones reused are without exception continuous-latent diffusion models, so the discrete-token cells can be filled only by training from scratch, the cost of Section 13, or by coupling token-based unimodal generators, a route no method we cover takes. Nor is every combination occupied—the empty and near-empty cells of Table 1 are themselves informative, and we return to them in Section 13. We treat each axis in turn.

4.1 Generation Strategy

The generation strategy is how the two modalities are produced relative to each other, that is, how the

dependency required by Problem 1 is realized in the computation graph.

4.1.1 Single-Tower

Formally $\theta_v = \theta_a = \theta_\times = \theta$: one network is applied to the concatenated sequence $[z_v; z_a]$, so every parameter sees both streams and the dependency is carried by the parameters themselves. A single network processes both modalities through shared parameters, with the two streams concatenated or interleaved into one sequence. Coupling is automatic, since every layer sees both modalities, at the cost of a representation that must serve both. Single-tower designs are the basis of AV-DiT (Wang et al., 2024b) and MM-LDM (Sun et al., 2024), and of recent open systems such as MOVA (SII-OpenMOSS Team, 2026), Apollo (formerly Klear) (Wang et al., 2026), and 3MDiT (Li et al., 2025).

4.1.2 Dual-Tower

Formally $\theta = (\theta_v, \theta_a, \theta_\times)$ with $\theta_v \cap \theta_a = \emptyset$: each stream is processed by its own tower, and information is exchanged only through the cross-modal parameters $\theta_\times$, so the dependency between v and a is carried by $\theta_\times$ alone. Two modality-specific towers run in parallel and exchange information through cross-modal connections such as cross-attention or bridge layers. The towers retain modality-specific inductive biases while the connections carry the dependency between v and a . This is the most common choice in the literature, adopted by MM-Diffusion (Ruan et al., 2023), JavisDiT (Liu et al., 2026a), BridgeDiT (Guan et al., 2025), Ovi (Low et al., 2025), UniAVGen (Zhang et al., 2025), and LTX-2 (HaCohen et al., 2026), among others.

4.1.3 Cascaded

Formally $p_\theta(v, a \mid c) = p_{\theta_1}(v \mid c) p_{\theta_2}(a \mid v, c)$, or the symmetric factorization, with disjoint θ_1, θ_2 and sequential sampling; the coupling is carried by the conditioning path rather than by shared parameters. The modalities are produced in sequence, with the second conditioned on the first, which reduces joint generation to a generation step followed by a cross-modal step. Movie Gen (Polyak et al., 2024) is the representative instance, generating video from text and then audio from the generated video.

4.1.4 Unified-Token

Formally ϕ_v, ϕ_a are quantizers into finite vocabularies, the pair is serialized into a single sequence $s = \pi(z_v, z_a)$, and $p_\theta(s) = \prod_t p_\theta(s_t \mid s_{<t})$ or a masked variant. Both modalities are discretized into tokens and modeled as a single sequence by an autoregressive or masked transformer, so the dependency is captured by the sequence model and the same backbone can serve multiple tasks by reordering inputs and outputs. No joint method covered here adopts this strategy. UniForm (Zhao et al., 2025) comes closest—it serializes the two modalities into one sequence and shares a denoiser across tasks distinguished by task tokens—but it does so over continuous VAE latents with a diffusion objective rather than over discrete vocabularies, which places it in the single-tower category. The unified-token cell is thus unoccupied among joint models, in contrast to the unimodal literature where token-based video and audio generation are established, an asymmetry we return to in Section 13.

4.1.5 Guidance-Based

Formally the two frozen models are coupled only in the sampler Γ , for example by adjusting the score of the pair $z = (z_v, z_a)$ as $\nabla_z \log p_\theta(z \mid c) + \lambda \nabla_z \mathcal{S}(\psi_v(z_v), \psi_a(z_a))$ with guidance weight λ ; θ is never updated jointly. Two pretrained unimodal models are frozen and coupled only at inference, through a guidance term such as a classifier or an alignment score that nudges the two samples toward mutual consistency. No joint training is required, which trades fidelity for flexibility. Seeing-and-Hearing (Xing et al., 2024a) couples two frozen generators through an Image-Bind (Girdhar et al., 2023) alignment score, and MMDisCo (Hayakawa et al., 2025) through a trained discriminator applied as sampling guidance.

4.2 Audio Representation

The audio representation is the realization of ϕ_a in Definition 2, and it fixes the space in which audio is generated. We discuss each choice in turn.

4.2.1 Continuous Latent

A neural audio autoencoder maps the waveform to a compact continuous latent in which a diffusion or flow model is trained. Writing the encoder as ϕ_a and its decoder as ψ_a (Table 2), the autoencoder is trained so that

$$z_a = \phi_a(a) \in \mathbb{R}^{T_z \times d}, \quad \psi_a(z_a) \approx a, \quad (1)$$

with a reconstruction loss plus, in the variational form (Kingma and Welling, 2014), a KL regularizer on the encoder’s posterior over z_a ; here $T_z \ll T_a$ is the compressed length and d the channel width. Generation then follows latent diffusion (Ho et al., 2020; Rombach et al., 2022): at diffusion step u , noised latents $z_u = \sqrt{\bar{\alpha}_u} z_a + \sqrt{1 - \bar{\alpha}_u} \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ are drawn along a noise schedule $\bar{\alpha}_u$, a network ϵ_θ is trained to minimize

$$\mathcal{L} = \mathbb{E}_{z_a, u, \epsilon} \|\epsilon - \epsilon_\theta(z_u, u, c)\|_2^2, \quad (2)$$

and sampling denoises from pure noise to $\hat{z}_a$, decoded as $\hat{a} = \psi_a(\hat{z}_a)$. This is the dominant choice in recent joint models: it is used by twenty-eight of the twenty-nine methods in Table 1, spanning early dual-tower models such as CoDi (Tang et al., 2023) through recent systems including LTX-2 (HaCohen et al., 2026) and MOVA (SII-OpenMOSS Team, 2026).

4.2.2 Discrete Tokens

A neural codec instead quantizes the latent: a codebook $\mathcal{C} = \{e_1, \dots, e_K\} \subset \mathbb{R}^d$ replaces each latent frame $z_a^{(i)}$ by its nearest entry,

$$q(z_a^{(i)}) = e_{k^*}, \quad k^* = \arg \min_k \|z_a^{(i)} - e_k\|_2, \quad (3)$$

trained with codebook and commitment terms $\|\text{sg}[z_a^{(i)}] - e_{k^*}\|_2^2 + \beta \|z_a^{(i)} - \text{sg}[e_{k^*}]\|_2^2$ under a straight-through gradient, where $\text{sg}[\cdot]$ denotes stop-gradient (van den Oord et al., 2017); practical audio codecs quantize residually over a stack of such codebooks, typically replacing the codebook term with an exponential-moving-average update of the selected entries (Défossez et al., 2022). The resulting index sequence supports autoregressive or masked modeling and a shared treatment with tokenized video. No joint method covered

here, however, generates audio as codec tokens—UniForm (Zhao et al., 2025) serializes the modalities into one sequence but over continuous latents (§4.2.1)—leaving this cell, like the waveform, unoccupied.

4.2.3 Mel-Spectrogram

Audio is represented as a mel spectrogram, obtained from the short-time Fourier transform through a mel filterbank M ,

$$m = \log(M |\text{STFT}(a)|^2) \in \mathbb{R}^{F \times T_m}, \quad (4)$$

with F mel bins and T_m spectral frames, and treated as an image-like array, which makes image generative machinery directly applicable but requires a separate vocoder to recover the waveform from a generated $\hat{n}$. MM-Diffusion (Ruan et al., 2023) is the representative instance among joint models.

4.2.4 Waveform

The model operates on the raw waveform $a \in \mathbb{R}^{T_a}$ directly, preserving full fidelity at the cost of modeling a very long and densely sampled sequence. No method we cover generates audio directly in the waveform domain, a consequence of the sequence lengths involved at audio sampling rates, an unoccupied corner we return to in Section 13.

4.3 Video Representation

The video representation is the realization of ϕ_v , with the same fidelity, cost, and compatibility trade-offs; the constructions mirror Equations 1–3 with v in place of a , and we discuss each choice in turn.

4.3.1 3D-VAE

A 3D autoencoder compresses space and time jointly,

$$z_v = \phi_v(v) \in \mathbb{R}^{T_v/s_t \times H/s_s \times W/s_s \times d_v}, \quad (5)$$

with temporal stride s_t , spatial stride s_s , and channel width d_v , producing a spatio-temporal latent in which a single backbone models the whole clip under the objective of Equation 2. This is the dominant choice in recent video and joint models. Nineteen of the methods in Table 1 use it, including Movie Gen (Polyak et al., 2024), JavisDiT (Liu et al., 2026a), Ovi (Low et al., 2025), LTX-2 (HaCohen et al., 2026), Apollo (Wang et al., 2026), and—through the pretrained Open-Sora autoencoder—UniForm (Zhao et al., 2025).

4.3.2 2D-VAE plus Temporal

A per-frame 2D autoencoder compresses each frame independently, $z_v^{(t)} = \phi_{v,2D}(v^{(t)})$ for $t = 1, \dots, T_v$, and a separate temporal module models motion across the stacked latent frames $(z_v^{(1)}, \dots, z_v^{(T_v)})$. CoDi (Tang et al., 2023), AV-DiT (Wang et al., 2024b), SVG (Ishii et al., 2024), and Animate-and-Sound (Wang et al., 2025c) take this route.

4.3.3 Discrete Tokens

Video is quantized into discrete tokens by the construction of Equation 3 applied to spatio-temporal patches, enabling autoregressive or masked modeling and a shared sequence with tokenized audio. No joint method covered here takes this route—UniForm (Zhao et al., 2025) fuses the streams as continuous latent tokens (§4.3.1)—mirroring the unoccupied discrete-audio cell.

4.3.4 Pixel

The model operates directly on pixels $v \in \mathbb{R}^{T_v \times H \times W \times 3}$, with no learned compression of the visual stream. MM-Diffusion (Ruan et al., 2023) is the sole pixel-space instance, reflecting the resolutions feasible when the work appeared.

4.4 Alignment Enforcement

Alignment enforcement is how a method raises the alignment score $\mathcal{S}$ of Definition 1, that is, the stage of the pipeline at which correspondence between the two streams is imposed (see Section 8).

4.4.1 Cross-Attention

Cross-attention layers let the audio stream attend to the video stream and vice versa, carrying alignment information between the two towers or branches—an architectural mechanism. It is the dominant mechanism, used by nineteen methods including MM-Diffusion (Ruan et al., 2023), BridgeDiT (Guan et al., 2025), Ovi (Low et al., 2025), and OmniForcing (Su et al., 2026).

4.4.2 Shared Positional Encoding

A shared positional or rotary encoding (Su et al., 2021) ties the two streams to a common time axis, so that tokens at the same physical time are forced into correspondence at the level of position—likewise an architectural mechanism. ALIVE (Guo et al., 2026b) and JavisDiT++ (Liu et al., 2026b) use temporally aligned rotary encodings for frame-level correspondence; AV-DiT (Wang et al., 2024b),

Apollo (Wang et al., 2026), and 3MDiT (Li et al., 2025) share positional information across streams.

4.4.3 Discriminator

An auxiliary joint discriminator is trained to score whether a pair is jointly real, and its gradient is applied as guidance at sampling time while the base generators stay frozen—an inference-time mechanism rather than a training objective for the generators. MMDisCo (Hayakawa et al., 2025), a trained discriminator applied as sampling guidance, is the only instance we cover.

4.4.4 Classifier Guidance

An external classifier or alignment model guides sampling toward consistent pairs at inference, in the manner of classifier guidance for diffusion models (Dhariwal and Nichol, 2021), without changing the generator’s weights—an inference-time mechanism. Seeing-and-Hearing (Xing et al., 2024a) is the representative case.

4.4.5 Explicit Prior

A separately estimated spatio-temporal prior is injected to align the streams, decoupling the synchronization signal from the generators—an architectural or an external mechanism, depending on whether the prior is learned jointly or estimated separately. JavisDiT (Liu et al., 2026a) estimates a hierarchical spatio-temporal prior for this purpose, retained in JavisDiT++ (Liu et al., 2026b).

4.5 Pretraining Reuse

The final axis is the source of the model’s weights, which strongly affects data and compute cost (see Section 13).

4.5.1 From-Scratch

The model is trained from random initialization on paired audio-visual data. Twelve methods train this way, including MM-Diffusion (Ruan et al., 2023), Movie Gen (Polyak et al., 2024), JavisDiT (Liu et al., 2026a), Ovi (Low et al., 2025), LTX-2 (Ha-Cohen et al., 2026), and MOVA (SII-OpenMOSS Team, 2026).

4.5.2 Single Pretrained

One pretrained backbone, typically a video model, is reused and the other modality is added on top. ALIVE (Guo et al., 2026b), Hallo-Live (Li et al., 2026), and OmniForcing (Su et al., 2026) build on one pretrained backbone.

4.5.3 Dual Pretrained

Two pretrained backbones, one per modality, are reused and coupled, so that joint training only learns the connections. CoDi (Tang et al., 2023), Seeing-and-Hearing (Xing et al., 2024a), SVG (Ishii et al., 2024), BridgeDiT (Guan et al., 2025), and CCL (Ma et al., 2026) couple two pretrained models.

5 Joint Audio-Visual Editing

Joint editing outputs a modified pair, $(v', a') \sim p_{\theta}(v', a' \mid v, a, e)$ (Problem 3), propagating the edit across modalities while preserving untouched content.

5.1 The Space of Audio-Visual Edits

This section mirrors Table 4 one to one: nine categories of edit types with representative operations and use cases (rendered radially in Figure 3). One-stream edits retain the other stream unchanged. Development concentrates in joint content, synchronization, and text-driven control; the empty cells are the research agenda.

5.2 Synchronization

Synchronization edits alter timing rather than content: lip-sync correction, alignment, re-timing. Dubbing is the developed instance: EdiDub (Manela et al., 2025) re-synchronizes lips by content-aware mouth-region editing, JUST-DUB-IT (Chen et al., 2026) jointly generates translated audio and synchronized facial motion via a low-rank adapter, and EditYourself (Flynn et al., 2026) targets identity-preserving re-voicing; alignment and re-timing have no dedicated methods.

5.3 Joint Content

Insertion, removal, and replacement in both streams is the prototypical joint edit—a passing car arrives with its engine sound. AV-Edit (Guo et al., 2026a) gates a masked-autoencoder-plus-DiT pipeline by audio-visual correlation; Object-AVEdit (Fu et al., 2025) reaches the same operations by inversion and regeneration. Still object-scoped: scene swaps and ambience-level insertions remain undemonstrated.

5.4 Cross-Modal Transfer

Deriving one stream from the other—re-voicing a portrait, re-soundtracking an edited video, stylizing both to a reference—runs the problems of Section 7 inside an editing pipeline; editing-specific is



Figure 3: **The edit-type taxonomy as a wheel.** The nine categories of Table 4 with their edit types arranged radially, in the same order and colors as the table; sector size is proportional to the number of edit types. Representative operations and example use cases for each type are enumerated in Table 4.

propagation, an edit in one modality inducing the consistent change in the other automatically, and the bidirectional type is unrealized.

5.5 Identity & Performance

Coupled face-and-voice swaps, performance transfer, and joint emotion edits are reached today only through dubbing (JUST-DUB-IT (Chen et al., 2026), EditYourself (Flynn et al., 2026)); the uni-modal ingredients are mature, missing only the coupling that keeps a swapped face and converted voice the same person (cf. §13).

5.6 Scene & Environment

Relocation, time-of-day and weather shifts, crowd changes, and acoustic coupling (reverberation inferred from depicted geometry) require audio edits proportional to the visual change; no method covered here targets these, the nearest machinery being ambience generation (§7.1).

5.7 Narrative

Cut-level edits—J- and L-cut splicing, reordering, summarization—are what professional editors do most, and no model covered here supports them: current methods edit within a shot, while narrative editing reasons across shots and modalities at once—the clearest open opportunity the taxonomy

Table 4: **A taxonomy of audio-visual edits:** nine categories broken into 28 edit types, each with representative edits and an example use case. Each type is additionally annotated by its dominant modality coupling: **A+V** denotes a genuinely joint edit that requires reasoning over both modalities; **V→A** denotes a video-driven edit with an audio consequence (or audio derived from video); **A→V** denotes the reverse; **V** and **A** denote edits that are primarily single-modality, with the other modality passive or unchanged. Categories are color-coded for clarity.

Category	Edit/Gen. Type	Modality	Representative Edits	Example Use Case
Synchronization	Lip Sync	A+V	Lip sync correction, dubbing alignment, viseme generation	Aligning dubbed audio to mouth motion
	AV Alignment	A+V	Foley alignment, beat alignment, event sync	Matching footsteps to visual steps
	Re-timing	A+V	Joint time stretch, slow motion, speed ramping	Slowing a scene with pitch-preserved audio
Joint Content	Joint Insertion	A+V	Object with sound, character with voice, ambience addition	Adding a passing car with engine noise
	Joint Removal	A+V	Object removal with sound suppression, character removal	Removing a person and their voice
	Joint Replacement	A+V	Object swap, scene swap, character swap	Replacing a dog with a cat (visual + sound)
Cross-Modal Transfer	Audio to Video	A→V	Talking head from speech, music-driven video	Animating a portrait from voice
	Video to Audio	V→A	Foley from video, ambience from scene, music from mood	Generating sound effects from silent video
	Bidirectional	A+V	AV style transfer, joint enhancement, coupled denoising	Stylizing both modalities to a reference
Identity & Perf.	Character Identity	A+V	Face swap with voice swap, aging, full replacement	Replacing an actor in face and voice
	Performance Transfer	A+V	Facial reenactment, gesture transfer, puppetry	Driving an avatar with a real performance
	Expression	A+V	Emotion editing across face and voice, energy, persona	Making a sad scene appear joyful
Scene & Environment	Setting	A+V	Location transfer, time-of-day, weather	Changing day to night with night ambience
	Acoustic Coupling	V→A	Reverb to visual space, occlusion, acoustic shadows	Matching reverb to a depicted cathedral
	Crowd & Background	A+V	Crowd density with noise, traffic with engines	Adding a crowd with crowd noise
Narrative	Cut & Splice	A+V	AV cut detection, J-cuts, L-cuts, montage	Editing dialogue with overlapping audio
	Reordering	A+V	Scene reordering, dialogue reordering, chronology	Rearranging scenes while preserving audio
	Length	A+V	Summarization, expansion, highlight extraction	Producing a 30s highlight from a long clip
Generative	Text-to-AV	A+V	Text-to-video with audio, music video, talking head	Generating a music video from a prompt
	AV Inpainting	A+V	Masked region inpainting, occlusion, gap filling	Filling a missing segment in a clip
	AV Outpainting	A+V	Temporal extension, spatial extension, continuation	Extending a clip beyond its original duration
Quality & Restoration	Joint Restoration	A+V	Joint denoising, archival restoration	Restoring old film with audio
	Visual Restoration	V	Super-resolution, deblurring, color correction	Upscaling old footage
	Audio Restoration	A	Denoising, dereverberation, click removal	Cleaning a noisy dialogue track
	Sync Repair	A+V	Drift correction, lip sync repair, AV offset	Fixing audio that drifts out of sync
Cross-Cutting	Granularity	A+V	Frame, shot, scene, clip, with paired audio scales	Editing one frame vs the entire clip
	Control Modality	A+V	Text, reference clip, storyboard, parametric, trajectory	Prompting via natural language
	Consistency	A+V	Cross-modal, temporal, identity, stylistic	Maintaining identity across an edit

exposes.

5.8 Generative

Joint inpainting and outpainting sit on the editing-generation boundary: the machinery is that of Section 6, but fidelity to input binds outside the synthesized region; neither has a dedicated method in the literature we cover.

5.9 Quality & Restoration

Joint denoising, per-stream restoration, and drift repair raise fidelity while changing nothing else; independent restoration can break correspondence, and sync repair after such processing has no learned joint treatment—well posed, demanded, unclaimed.

5.10 Cross-Cutting

Granularity, control modality, and consistency cut across all categories; control differentiates current methods. The most general instruction is text: language-guided joint editing (Liang et al., 2024b) adapts a joint model to a single example so a textual edit propagates, and AvED (Lin et al., 2026) obtains it zero-shot by delta-denoising both streams under frozen unimodal models.

6 Joint Audio-Visual Generation

Joint generation outputs both streams, $(\hat{v}, \hat{a}) \sim p_\theta(v, a \mid c)$ (Problem 1); *joint* names what is generated, not how. Table 5 organizes the methods we cover, with the cross-modal and editing families, grouped by output.

6.1 Unconditional Joint Generation

The case $c = \emptyset$ exposes the dependency most directly; coupled diffusion over a paired latent, as in MM-Diffusion (Ruan et al., 2023), is the canonical instance.

6.2 Text-to-Audio-Visual Generation

Text-to-audio-visual generation samples $(\hat{v}, \hat{a}) \sim p_\theta(v, a \mid c_t)$ depicting the prompt in both streams. Methods span the taxonomy: dual-tower DiTs coupled through cross-attention (JavisDiT (Liu et al., 2026a), extended with modality-specific experts and aligned rotary encodings (Liu et al., 2026b)); cross-modal context learning (Ma et al., 2026); twin-backbone fusion (Ovi (Low et al., 2025)) and asymmetric interaction (UniAVGen (Zhang et al., 2025)) for lip sync and timbre; open systems scaling joint training (MOVA (SII-OpenMOSS Team, 2026), Apollo (Wang et al., 2026)); and partially documented systems (Wan 2.5 (Alibaba, 2025)), left unmarked on undisclosed axes in Table 1.

6.3 Image-to-Audio-Visual Generation

Conditioning on an image pins identity and layout, shifting the difficulty to motion and sound consistent with a fixed first frame; sharing an expert block between video and audio branches, as in Animate-and-Sound (Wang et al., 2025c), is representative.

6.4 Talking-Head and Speech-Driven Generation

Talking-head generation outputs a speaking face with its speech track from an identity image and

driving speech; correspondence reduces to single-frame lip synchronization. Hallo-Live (Li et al., 2026) reaches streaming avatars by attending to a short horizon of future phonetic cues.

6.5 Long-Form Joint Generation

Long-form generation adds coherence over minutes—errors accumulate, and identity, scene, and the audio-visual relationship must not drift; streaming formulations such as OmniForcing (Su et al., 2026) generate in causal blocks while distilling from a bidirectional teacher.

7 Cross-Modal Generation

Cross-modal generation outputs exactly one modality given the other (Problem 2): the input is observed and fixed, and the output must be made consistent with it.

7.1 Video-to-Audio Generation

Video-to-audio, $\hat{a} \sim p_\theta(a \mid v)$, is by far the most developed cross-modal setting: onsets must land at the exact frames of visual contact. One durable strategy learns correspondence before generating (Diff-Foley’s contrastive pretraining (Luo et al., 2023), STA-V2A (Ren et al., 2024b), VATT’s caption route (Liu et al., 2024d), V2A-Mapper’s frozen-model bridge (Wang et al., 2024a)); a second adopts flow matching for faster sampling and tighter synchronization (Frieren (Wang et al., 2024d), VAFlow (Wang et al., 2025b), Foley-Flow (Mo and Song, 2025), MMAudio (Cheng et al., 2025)), with masked and causal token models alongside (MaskVAT (Pascual et al., 2024), SoundReactor (Saito et al., 2025)); a third attaches control to a fixed generator (FoleyCrafter (Zhang et al., 2026), Video-Foley (Lee et al., 2025), Mel-QCD (Wang et al., 2025a), MultiFoley (Chen et al., 2025), TARO (Ton et al., 2025)). Control has lately moved to instructions (ThinkSound (Liu et al., 2025), Hear-Your-Click (Liang et al., 2025), SelVA (Lee et al., 2026)); Foley-Omni (Tao et al., 2026) folds speech, effects, and music into one generator, and closed systems such as Google’s V2A (Google DeepMind, 2024) disclose little.

7.2 Audio-to-Video Generation

Audio-to-video, $\hat{v} \sim p_\theta(v \mid a)$, is structurally harder: one track licenses many videos. Dedicated attempts are isolated (Yariv et al. (2024) adapt a frozen text-to-video model and introduce

Table 5: **Taxonomy of methods for joint audio-video generation and editing.** Approaches are categorized by their *output*: the first group outputs both modalities together, V+A (Sec. 6); the second outputs a single modality conditioned on the other, A from V or V from A (Sec. 7); and the third outputs a modified version of an existing pair, V'+A' (Sec. 5). **Task:** Uncond. = unconditional joint generation, T2AV = text-to-audio-video, I2AV = image-to-audio-video, Any2AV = any-modality input, V2A = video-to-audio, A2V = audio-to-video, AVE = joint audio-video editing, Dub = joint dubbing. **Inputs:** T (text), V (video), A (audio), I (image), A-ref (reference audio), instr. (instruction), trans. (translated transcript). **Output:** V+A jointly generated, V'+A' jointly edited. **Backbone:** UNet, DiT (diffusion transformer (Peebles and Xie, 2023)), Flow (rectified flow / flow matching). **Train:** ZS (zero-shot), OS (one-shot), FT (fine-tune), SC (from-scratch), Adp (adapter-only), LoRA. **Code:** ✓ open-source, ✗ not released. – marks entries not applicable or not publicly disclosed (Google V2A, Wan 2.5).

Method	Task	Inputs	Output	Backbone	Architecture	Conditioning	Alignment	Train	Code
<i>Joint Audio-Video Generation (Sec. 6) — output both streams: $(\hat{v}, \hat{a}) \sim p_{\theta}(v, a c)$</i>									
MM-Diffusion (Ruan et al., 2023)	Uncond.	–	V+A	Coupled UNet	Sequential dual UNet	Random-shift cross-attn	Joint denoising	SC	✓
CoDi (Tang et al., 2023)	Any2AV	T, V, A, I	V+A	Latent UNet	Composable diffusion	Bridging encoders	Cross-modal latents	Adp	✓
Seeing-and-Hearing (Xing et al., 2024a)	T2AV	T, V, A	V+A	Frozen UNets	Two single-modal models	ImageBind aligner	Latent classifier guidance	ZS	✓
AV-DiT (Wang et al., 2024b)	T2AV	T	V+A	Shared DiT	Single DiT, two heads	Lightweight adapters	Shared self-attn	FT	✗
MM-LDM (Sun et al., 2024)	T2AV	T	V+A	Latent UNet	Hierarchical latent	Hierarchical multi-modal	Shared latent	SC	✗
Movie Gen (Polyak et al., 2024)	T2AV	T, I	V+A	DiT (Flow)	Cascaded T2V → V2A	Cascaded conditioning	Cascaded	SC	✗
SVG (Ishii et al., 2024)	T2AV	T	V+A	Two pretrained DiTs	Adapted dual-tower	Lightweight bridging	Cross-modal exchange	FT	✗
MMDiSCo (Hayakawa et al., 2025)	T2AV	T	V+A	Two UNets	Frozen + joint discriminator	Discriminator guidance	Adversarial alignment	FT	✓
SyncFlow (Liu et al., 2024b)	T2AV	T	V+A	Dual DiT (Flow)	d-DiT, decoupled multi-stage	Text on both branches	Joint fine-tune	SC	✗
JavisDiT (Liu et al., 2026a)	T2AV	T	V+A	Joint DiT	AV-DiT with ST cross-attn	HiST-Syppo prior	Hierarchical ST attn	SC	✓
JavisDiT++ (Liu et al., 2026b)	T2AV	T	V+A	Joint DiT (MS-MoE)	Dual-branch with MS-MoE	HiST prior + TA-RoPE	Frame-level TA-RoPE, AV-DPO	SC	✓
BridgeDiT (Guan et al., 2025)	T2AV	T	V+A	DiT	Dual-tower with bridge	Decoupled T_V / T_A captions	Bidirectional bridge	FT	✓
ALIVE (Guo et al., 2026b)	T2AV	T, I	V+A	DiT	Dual+single stream	TA-CrossAttn + UniTemp-RoPE	Strict temporal RoPE	FT	✗
Ovi (Low et al., 2025)	T2AV	T	V+A	Twin DiT	Twin backbones, cross fusion	Cross-modal fusion	Symmetric fusion	SC	✓
UniAVGen (Zhang et al., 2025)	T2AV	T	V+A	Joint DiT	Dual-branch parallel DiT	Asym. cross-modal interaction	Face-aware modulation, MA-CFG	SC	✓
Animate-and-Sound (Wang et al., 2025c)	I2AV	I	V+A	Dual-tower DiT	Decompose + expert blocks	Image-conditioned	Mutual influence	FT	✗
CCL (Ma et al., 2026)	T2AV	T	V+A	Dual-stream DiT	Cross-modal context learning	Decoupled cross-modal context	Context alignment	FT	✗
Hallo-Live (Li et al., 2026)	I2AV	I, A	V+A	Dual-stream DiT	Async. dual-stream, streaming	Future-expanding attn	Streaming lip-sync, HP-DMD	FT	✓
UniForm (Zhao et al., 2025)	Any2AV	T, V, A	V+A	Multi-task DiT	Shared denoiser, task tokens	Task tokens	Shared latent	SC	✗
Wan 2.5 (Alibab, 2025)	T2AV	T, I, A	V+A	DiT	Multilingual joint	T5 + audio + image	–	–	✗
LTX-2 (HaCohen et al., 2026)	T2AV	T	V+A	Asym. dual DiT (14B+5B)	Bidir. cross-attn	Modality-CFG, AdaLN	Bidir. cross-attn	SC	✓
MOVA (SII-OpenMOSS Team, 2026)	T2AV	T	V+A	DiT	Open joint model	Multi-track conditioning	End-to-end joint	SC	✓
Apollo (Wang et al., 2026)	T2AV	T	V+A	Single-tower MM-DiT	Omni-Full Attention	Progressive multi-task	Tight AV alignment	SC	✗
3MDiT (Li et al., 2025)	T2AV	T	V+A	Tri-modal DiT	Isomorphic A/V branches	Trimodal omni-blocks	Dynamic text + AV co-evolve	SC/FT	✗
OmniForcing (Su et al., 2026)	T2AV	T	V+A	Streaming DiT	Causal AR distilled from LTX-2	Distilled bidirectional	Streaming sync	FT	✓

Table 5: Taxonomy of methods for joint audio-video generation and editing (continued).

Method	Task	Inputs	Output	Backbone	Architecture	Conditioning	Alignment	Train	Code
Cross-Modal Generation (Sec. 7) — output one stream given the other: $\hat{a} \sim p_{\theta}(a v)$ or $\hat{v} \sim p_{\theta}(v a)$									
Diff-Foley (Luo et al., 2023)	V2A	V	A	Latent UNet	CAVP + latent diffusion	—	Contrastive AV pretraining	SC	✓
Foley Analogies (Du et al., 2023)	V2A	V/A-ref	A	Latent diffusion	Reference-conditioned Foley	Audio reference	Onset transfer from exemplar	SC	✓
V2A-Mapper (Wang et al., 2024a)	V2A	V	A	Frozen foundation	Lightweight vision-audio mapper	—	Foundation-model embedding match	Adp	✗
Video-Foley (Lee et al., 2025)	V2A	V/T,A-ref	A	Latent UNet	RMS two-stage control	Text + audio ref	RMS envelope conditioning	Adp	✓
FoleyCrafter (Zhang et al., 2026)	V2A	V/T	A	Latent UNet	Semantic adapter + temporal ctrl.	Text	Temporal controller	Adp	✓
Frieren (Wang et al., 2024d)	V2A	V	A	Flow (RF)	Rectified flow matching	—	Onset-aligned flow	FT	✓
MaskVAT (Pascual et al., 2024)	V2A	V	A	Masked transformer	Masked generative transformer	—	Enhanced synchronicity	SC	✗
STA-V2A (Ren et al., 2024b)	V2A	V/T	A	Latent UNet	Local + global visual features	Text	Semantic + temporal alignment	FT	✓
VATT (Liu et al., 2024d)	V2A	V/T	A	Latent UNet	Caption-mediated generation	Text	Caption-level semantic match	SC/LoRA	✓
Google V2A (Google DeepMind, 2024)	V2A	V/T	A	Latent diffusion	Prompt-conditioned diffusion	Text	Onset conditioning	—	✗
MMAudio (Cheng et al., 2025)	V2A	V/T	A	Flow (RF)	Multimodal joint training	Text	Dedicated synchronization module	SC	✓
Mel-QCD (Wang et al., 2025a)	V2A	V/T	A	Latent UNet	Mel decomposition + ControlNet	Text	Mel quantization-continuum	Adp	✓
VAFLOW (Wang et al., 2025b)	V2A	V	A	Flow (RF)	Cross-modality flow matching	—	Cross-modal flow coupling	SC	✗
Foley-Flow (Mo and Song, 2025)	V2A	V	A	Flow (RF)	Masked AV align + dynamic flow	—	Masked audio-visual alignment	SC	✗
MultiFoley (Chen et al., 2025)	V2A	V/T,A-ref	A	DiT	Multi-conditional training	Text + audio ref	Onset alignment	SC	✗
TARO (Ton et al., 2025)	V2A	V	A	DiT	Time-step-adaptive repr. alignment	—	Onset-aware conditioning	SC	✓
ThinkSound (Liu et al., 2025)	V2A	V/T,mask	A	DiT	MLLM chain-of-thought	Text + click/mask	Reasoned event placement	FT	✓
Hear-Your-Click (Liang et al., 2025)	V2A	V/T,click	A	DiT	Object-centric conditioning	Click/mask	Object-level onset	FT	✓
SeIVA (Lee et al., 2026)	V2A	V/T	A	DiT	Text-conditioned selective V2A	Text	Selective source onset	FT	✓
SoundReactor (Saito et al., 2025)	V2A	V	A	Causal AR + diff. head	Frame-level online generation	—	Streaming frame-level sync	SC	✓
Foley-Omni (Tao et al., 2026)	V2A	V/T	A	DiT	Unified task $\rightarrow$ full soundtrack	Text	Multi-track soundtrack alignment	SC	✓
AV-Link (Haji-Ali et al., 2025)	V2A/A2V	V or A	A or V	Flow (frozen)	Frozen backbones + feature links	Text + audio ref	Temporally-aligned diff. features	Adp	✗
Joint Audio-Video Editing (Sec. 5) — output a modified pair: $(v', a') \sim p_{\theta}(v', a' v, a, e)$									
Lang.-Guided AV Edit (Liang et al., 2024b)	AVE	V/A,T	V' + A'	Joint AV diffusion	One-shot LoRA adaptation	Text + paired AV	Cross-modal sem. enhancement	OS	✗
AvED (Lin et al., 2026)	AVE	V/A,T	V' + A'	Frozen latent UNets	Cross-modal delta denoising	Text prompts	Patch-level AV delta align.	ZS	✓
EdiDub (Manela et al., 2025)	Dub	V/A,mask	V'	3D UNet (diff.)	Two-stage content-aware edit	Quantized HubBERT audio	AdaLN audio modulation	SC	✗
Object-AVEdit (Fu et al., 2025)	AVE	V/A,T	V' + A'	Mochi-1 + audio DiT	Inversion-regeneration	Source/target prompts	—	SC/ZS	✗
AV-Edit (Guo et al., 2026a)	AVE	V/A,T,instr.	A'	MM-DiT	CAV-MAE-Edit + MM-DiT	AV semantic control	AV correlation gating	FT	✗
JUST-DUB-IT (Chen et al., 2026)	Dub	V/A,T-trans.	V' + A'	Joint AV diffusion	LoRA on AV foundation	Audio + video joint cond.	Joint AV prior	LoRA	✓
EditYourself (Flynn et al., 2026)	AVE	V/A,script	V'	DiT	Audio-conditioned V2V	Audio + region masks	Identity-preserving lip-sync	FT	✗

AV-Align); otherwise the direction is supported only incidentally, by MM-Diffusion’s joint distribution (Ruan et al., 2023), Seeing-and-Hearing’s direction-indifferent aligner (Xing et al., 2024a), UniAVGen’s task list (Zhang et al., 2025), and AV-Link’s bidirectional linking (Haji-Ali et al., 2025); even video-to-music (§8.4) runs almost entirely the other way (§13).

7.3 Speech-to-Video Generation

Speech-to-video outputs only the visual stream from speech and an identity image; correspondence reduces to the viseme-phoneme match. Portrait animation dominates—speech-to-gesture (Ginosar et al., 2019), animators predicting 3D coefficients (Zhang et al., 2023a) or denoising video directly (Tian et al., 2024; Xu et al., 2024a), extended to full figures (Corona et al., 2025; Lin et al., 2025)—while beyond the portrait the mapping is radically one-to-many and largely untouched (§13).

7.4 Foley and Sound-Effect Generation from Video

Foley narrows video-to-audio to the diegetic sound of visible actions, where timing is tightest—a footstep a few frames off is wrong, not degraded—and where practice demands control over which sources sound, when, and how loud: exemplar transfer (Du et al., 2023), envelope control (Lee et al., 2025), multi-signal conditioning (Chen et al., 2025), and user selection (Liang et al., 2025; Lee et al., 2026).

8 Alignment and Synchronization

Alignment is the property that ties together every setting in this work, and we treat it once here rather than repeating it in each section. It is the operational form of Definition 1: a pair (v, a) is aligned when embeddings from a contrastively trained encoder pair $f_v : \mathcal{V} \rightarrow \mathbb{R}^{d_e}$ and $f_a : \mathcal{A} \rightarrow \mathbb{R}^{d_e}$ —distinct from the generative encoders (ϕ_v, ϕ_a) of Definition 2—are close under cosine similarity, either globally or per time step.

Definition 3 (Temporal synchronization). A pair (v, a) is temporally synchronized at tolerance δ when, for each time index t , the visual event at frame $v^{(t)}$ corresponds to an audio event within $[t - \delta, t + \delta]$ of a ; lip sync (mouth shape vs. phoneme) and beat alignment (motion peak vs. beat) are its special cases.

Definition 3 makes precise the temporal component of correspondence (Definition 1); the score

S itself is operationalized by the learned encoders above, and the subsections below organize methods by the kind of event they align and the tolerance they target.

8.1 Temporal Synchronization

General temporal synchronization places arbitrary acoustic events at the frames of their visual cause, the requirement underlying video-to-audio and joint generation alike, and the tolerance δ of Definition 3 that a method achieves is the primary measure of its temporal quality. Methods reach it in three broadly different places. Some supply alignment through a representation learned in advance, as in the contrastive audio-visual pretraining of Diff-Foley (Luo et al., 2023), so that the generator inherits correspondence rather than enforcing it. Others add machinery dedicated to timing: MMAudio (Cheng et al., 2025) attaches an explicit synchronization module, TARO (Ton et al., 2025) conditions on onsets while adapting representation alignment across timesteps, and MaskVAT (Pascual et al., 2024) targets synchronicity directly in a masked token model. A third group builds it into the architecture, either through cross-attention between the two streams—the dominant choice, adopted by nineteen of the methods in Table 1—or through a shared temporal encoding that forces tokens at the same physical time into correspondence, as in JavisDiT++ (Liu et al., 2026b) and ALIVE (Guo et al., 2026b). Approaches that impose alignment only at inference, whether by discriminator (Hayakawa et al., 2025) or by classifier guidance (Xing et al., 2024a), are now the exception, which is itself evidence that synchronization has migrated from a post-hoc correction into the model.

8.2 Semantic Alignment

Semantic alignment is the weaker, global property that the sources in v and a match in identity even when their timing is loose. It is necessary but not sufficient for correspondence: a clip whose sources agree but whose events are misplaced still feels out of step. In practice it is operationalized through contrastive audio-visual embeddings, which several methods reuse directly as a guidance term—Seeing-and-Hearing (Xing et al., 2024a) steers two frozen unimodal generators toward agreement using an ImageBind (Girdhar et al., 2023) aligner—or as a training signal. An alternative is to route the alignment through language: VATT (Liu et al., 2024d)

captions the video and generates audio from the caption, which makes the semantic link explicit and controllable at the cost of the temporal precision that a direct visual conditioning path preserves. Recent benchmarks suggest this is where current models are weakest, with AVGen-Bench (Zhou et al., 2026) reporting a gap between strong audio-visual aesthetics and unreliable semantic grounding.

8.3 Lip-Sync and Phoneme-Level Alignment

Lip synchronization is the most demanding instance of Definition 3: the tolerance is on the order of a single frame, and viewers detect phoneme-to-viseme mismatch far more readily than any other misalignment. It is the binding constraint in talking-head generation, speech-to-video, and dubbing, and methods in those families are organized around it rather than merely evaluated on it. UniAVGen (Zhang et al., 2025) introduces face-aware modulation for exactly this purpose; Hallo-Live (Li et al., 2026) lets each generated video block attend to a short horizon of future phonetic cues so that streaming generation does not sacrifice lip accuracy; and in the editing setting, JUST-DUB-IT (Chen et al., 2026) and EditYourself (Flynn et al., 2026) must satisfy the same constraint while preserving speaker identity and the untouched regions of the source clip. Because human sensitivity here is unusually sharp, this is also the sub-problem with the most established automatic metrics, and the one where they agree best with human judgment.

8.4 Rhythmic and Beat-Level Alignment

For music, alignment is rhythmic rather than phonetic: motion peaks should coincide with musical beats. The relevant event is periodic, which makes the alignment both easier to measure and easier to violate in a way that is immediately noticeable. Most work in this setting runs from video to music rather than the reverse, generating a soundtrack whose beat structure follows observed motion. Early approaches tie note onsets to body movement in instrument performance (Gan et al., 2020; Su et al., 2020) and to human motion more generally (Gan et al., 2021), while later work targets background music for arbitrary video with explicit rhythmic control (Di et al., 2021; Zhuo et al., 2023) and extends the horizon over which rhythm must remain coherent (Yu et al., 2023). Dance video, where the motion is already organized around a beat, is the most constrained instance (Zhu et al., 2022), and the paired dance-and-music corpora

built for it (Li et al., 2021) are the standard evaluation setting. The reverse direction, generating video whose motion follows a given piece of music, remains comparatively unexplored, an instance of the broader asymmetry discussed in Section 7.2.

8.5 Consistency Across Long Sequences

Over long horizons, alignment must be maintained as well as achieved, since small per-step errors accumulate into visible and audible drift. This makes long-form coherence a distinct problem rather than an extension of short-clip synchronization, and it connects this section to the long-form generation problem of Section 6.5. Two families of solution have emerged. Streaming and causal formulations generate in blocks while distilling from a bidirectional teacher, as in OmniForcing (Su et al., 2026), or attach a diffusion head to a causal transformer to produce audio frame by frame under an online latency budget, as in SoundReactor (Saito et al., 2025). Alternatively, methods extend the generation window directly, whether through the long-form conditioning of MultiFoley (Chen et al., 2025) and Movie Gen’s audio branch (Polyak et al., 2024) or through architectures aimed at unbounded generation (Ergasti et al., 2025). Both remain evaluated on horizons far shorter than the minutes-long content the applications of Section 12 assume, which we return to in Section 13.

9 Architectures and Training Strategies

Having defined the tasks, we turn to the model families used to instantiate p_θ . A cascaded model factorizes the joint distribution as $p_\theta(v, a | c) = p_{\theta_1}(v | c) p_{\theta_2}(a | v, c)$, or the reverse, producing one modality first and the other conditioned on it (§4.1.3); a two-tower model instead parameterizes $p_\theta(v, a | c)$ jointly, denoising both streams in parallel with the coupling carried by $\theta_\times$ (§4.1.2). A unified backbone learns $p_\theta(v, a | c)$ directly with a single network that processes both modalities through shared parameters. Within any of these, diffusion and flow-matching approaches parameterize p_θ through a denoising process applied to v , a , or both, while autoregressive approaches tokenize the two modalities and model them as a single sequence. Architecture and task interact in predictable ways: cascaded models are common in cross-modal generation, where one modality is observed and the other is conditioned on it, whereas unified backbones are more common in joint gener-

ation, where both modalities are outputs of a shared distribution. Autoregressive discrete-token modeling, natural for streaming, remains unoccupied (§4.1).

9.1 Two-Tower and Cascaded Models

Two-tower and cascaded models keep the modalities in separate networks and couple them through cross-modal connections or through a generation order. They can reuse strong unimodal backbones and add only the coupling, which makes them data-efficient at the cost of a coordination burden between the towers.

9.2 Unified Multimodal Backbones

Unified backbones process both modalities with shared parameters, either as one fused sequence or as a single network with modality-specific heads. They capture the dependency between v and a most directly and are the basis of most recent joint models, at the cost of a representation that must serve both modalities at once.

9.3 Diffusion-Based Approaches

Diffusion and flow-matching approaches dominate both modalities. Joint variants apply the denoising process to a paired latent, with the coupling realized through shared layers or cross-attention, and they inherit the controllability of classifier-free guidance directly.

9.4 Autoregressive and Token-Based Approaches

Autoregressive and masked token models treat the two modalities as one sequence of discrete tokens, which makes streaming and variable-length generation natural and supports a single backbone across multiple tasks through input reordering.

9.5 Training Objectives and Losses

Beyond the per-modality reconstruction or denoising loss, joint methods add objectives that target correspondence directly, including contrastive alignment losses, adversarial joint-realism losses, and preference objectives that reward synchronization. The choice of objective is the training-time counterpart of the alignment-enforcement axis of Section 4.4.

10 Datasets and Benchmarks

Progress in the area is driven by the data used to train and evaluate the methods of the previous sec-

tions.

Definition 4 (Audio-visual dataset). A generation dataset is a collection $\mathcal{D} = \{(v_n, a_n, c_n)\}_{n=1}^N$ of paired video, audio, and optional conditioning signals, where c_n is typically a caption describing both modalities for joint generation and is empty for cross-modal generation. An editing dataset takes the richer form $\mathcal{D} = \{(v_n, a_n, e_n, v'_n, a'_n)\}_{n=1}^N$, where e_n is an edit instruction and (v'_n, a'_n) is the target pair.

10.1 Audio-Visual Generation Datasets

Generation datasets pair video with audio and, increasingly, with captions that describe both streams; Table 6 lists representative instances. The scale, domain, and caption quality of $\mathcal{D}$ bound what a model can learn about correspondence, and recent collections emphasize captions that describe the audio-visual relationship rather than either stream alone.

10.2 Audio-Visual Generation Benchmarks

Benchmarks fix a prompt set and an evaluation protocol so that methods can be compared on the same footing; Table 7 summarizes recent ones. Recent task-driven benchmarks such as AVGenBench (Zhou et al., 2026) evaluate text-to-audio-video generation at multiple granularities and expose a gap between strong audio-visual aesthetics and weak semantic reliability, while physically grounded benchmarks such as AV-Phys Bench (Cui et al., 2026) probe whether joint models respect the physics linking a visual event to its sound. No shared benchmark for joint audio-visual editing exists: each editing method covered here evaluates on data it assembled or repurposed itself, and none of these sets provides the (v, a, e, v', a') supervision that Definition 4 defines for an editing dataset—a gap that makes editing results mutually incomparable today.

11 Evaluation Metrics

Assessing the outputs of joint and cross-modal models requires measures that capture both per-modality quality and cross-modal consistency. For a generated video $\hat{v}$, a quality metric $Q_v(\hat{v})$, or $Q_v(\hat{v}, v)$ when a reference is available, scores visual fidelity. For a generated audio $\hat{a}$, a metric $Q_a(\hat{a})$ scores audio fidelity. For a generated pair, an alignment metric $A(\hat{v}, \hat{a})$ scores cross-modal

Table 6: **Representative datasets for joint and cross-modal audio-visual generation.** We group datasets by domain and report approximate scale and whether text captions are available (**Cap.**). The rightmost column lists the task each dataset most directly supports. Scales are approximate and refer to the commonly used release.

Dataset	Year	Domain	Approx. Scale	Cap.	Primary Task
AudioSet (Gemmeke et al., 2017)	2017	in-the-wild events	~2M clips, 10s each	✗	AV pretraining, V2A
VGGSound (Chen et al., 2020a)	2020	in-the-wild events	~200k clips, 10s each	✗	V2A, joint generation
Kinetics (Kay et al., 2017)	2017	human actions	~650k clips	✗	AV pretraining
Greatest Hits (Owens et al., 2016)	2016	object impacts	~1k videos	✗	Foley / impacts
MUSIC (Zhao et al., 2018)	2018	instrument solos/duets	714 videos	✗	music V2A, separation
URMP (Li et al., 2019)	2019	classical ensembles	44 multi-track pieces	✗	music V2A, separation
AIST++ (Li et al., 2021)	2021	dance with music	1408 seq., 1.1M frames	✗	music-to-motion/video
AVSpeech (Ephrat et al., 2018)	2018	talking faces	thousands of hours	✗	speech-driven, separation
VoxCeleb2 (Chung et al., 2018)	2018	talking faces	> 1M utterances	✗	talking-head, identity
TAVGBench (Mao et al., 2024)	2024	in-the-wild audible video	~1.7M clips, 11.8k h	✓	T2AV training and evaluation
MMTrail (Chi et al., 2024)	2024	trailers with music	> 20M clips (2M mm-captioned)	✓	music-video generation
JavisBench (Liu et al., 2026a)	2025	open-domain sounding video	10,140 captioned clips	✓	T2AV evaluation

Table 7: **Recent benchmarks for joint audio-visual generation.** Each benchmark fixes a prompt set and an evaluation protocol; the columns name the task targeted and the property tested.

Benchmark	Task	Property Tested
JavisBench (Liu et al., 2026a)	T2AV	quality and synchronization in diverse scenes
SAVGBench (Shimada et al., 2026)	joint gen.	spatial alignment between first-order-ambisonics audio and video
AVGen-Bench (Zhou et al., 2026)	T2AV	aesthetics vs. semantic reliability (text, speech, physics, music)
AV-Phys Bench (Cui et al., 2026)	joint gen.	physical commonsense across steady and transition scenes

consistency, which neither Q_v nor Q_a alone captures. Edited pairs require two further measures: a faithfulness metric that assesses whether the edit instruction e was applied, and a preservation metric that assesses whether content outside the edit region was left intact. Table 8 organizes the metrics in use by what they measure.

11.1 Video Quality and Fidelity

Visual quality metrics score the realism and prompt-faithfulness of $\hat{v}$, typically through distances between feature distributions of generated and real clips. They are necessary but insufficient, since a model can score well while ignoring the audio entirely.

11.2 Audio Quality and Fidelity

Audio quality metrics score the realism and prompt-faithfulness of $\hat{a}$, again through distributional distances or learned predictors. As with video, a high audio score does not imply correspondence with the visual stream.

11.3 Cross-Modal Alignment Metrics

Alignment metrics realize the score $\mathcal{S}$ of Definition 1, measuring either semantic agreement through contrastive audio-visual embeddings or temporal agreement through onset and beat distances. They are the metrics that distinguish joint

and cross-modal evaluation from unimodal evaluation.

11.4 Editing Faithfulness and Preservation

Editing metrics pair a faithfulness measure, which checks that the change named by e was applied, with a preservation measure, which checks that untouched regions are unchanged. The two are in tension, and reporting one without the other is misleading.

11.5 Human Evaluation Protocols

Because correspondence is ultimately a perceptual property, human evaluation remains the reference standard, typically through forced-choice comparisons on quality and synchronization. Automatic metrics are validated by their agreement with these judgments.

12 Applications

Joint and cross-modal models are deployed across a range of settings, each placing its own constraints on p_θ . Every application can be characterized by four elements: the input modalities, the output modalities, the underlying task of generation or editing, and the operational constraints such as latency, controllability, and identity preservation. Content creation, dubbing and accessibility, real-time avatars, and personalization each constrain

Table 8: **Evaluation metrics organized by what they measure.** Per-modality quality metrics score one stream in isolation; condition-alignment metrics score agreement with the input c ; cross-modal alignment metrics realize the score $\mathcal{S}$ of Def. 1; and editing metrics score the two competing requirements of Problem 3. **Modality** indicates the streams compared (V video, A audio, T text), and **Better** the preferred direction.

Metric	Measures	Modality	Better
<i>Per-modality quality</i>			
FID (Heusel et al., 2017), FVD (Unterthiner et al., 2018)	visual fidelity (distribution distance)	V	↓
Inception Score (Salimans et al., 2016)	visual quality and diversity	V	↑
FAD (Kilgour et al., 2019)	audio fidelity (distribution distance)	A	↓
KL (audio classifier)	audio semantic match	A	↓
<i>Condition alignment</i>			
CLIPScore (Radford et al., 2021; Hessel et al., 2021)	text-to-visual agreement	V/T	↑
CLAP score (Wu et al., 2023b)	text-to-audio agreement	A/T	↑
<i>Cross-modal alignment ($\mathcal{S}$, Def. 1)</i>			
ImageBind AV score (Girdhar et al., 2023)	audio-visual semantic agreement	V/A	↑
AV-Align (Yariv et al., 2024) / onset accuracy	temporal event synchronization	V/A	↑
LSE-C / LSE-D (SyncNet) (Chung and Zisserman, 2016)	lip-sync confidence / distance	V/A	↑ / ↓
Beat alignment	rhythmic synchronization	V/A	↑
<i>Editing (Problem 3)</i>			
Directional faithfulness	whether the edit e was applied	V / A	↑
Masked PSNR / LPIPS (Zhang et al., 2018)	preservation outside the edit region	V	↑ / ↓

p_θ differently—controllability, identity-preserving propagation, a hard latency budget, and subject consistency, respectively.

12.1 Film, Animation, and Content Creation

In content creation the priority is controllability and quality, and the task spans both joint generation of new clips and editing of existing footage, often with a soundtrack composed of speech, effects, and music together.

12.2 Dubbing, Translation, and Accessibility

Dubbing and translation are editing applications in which audio and video must change together while identity is preserved, which makes them the clearest instance of the propagation requirement of Problem 3. JUST-DUB-IT (Chen et al., 2026) generates translated speech and matching facial motion jointly while holding speaker identity fixed, and EditYourself (Flynn et al., 2026) addresses the related re-voicing case. Accessibility uses the same machinery in the other direction, adding or adapting content for different audiences, and shares the preservation constraint: source-preserving methods such as MMAudioSep (Takahashi et al., 2026) and the audio-follows-video-edit setting of CoherentAVEdit (Ishii et al., 2025) are the research counterparts. What distinguishes this family operationally is that a wrong edit is worse than no edit, since the input is real footage a user already has.

12.3 Virtual Avatars and Telepresence

Avatars and telepresence impose the only hard latency constraint in this work: generation must keep pace with speech, which rules out the bidirectional sampling that every other application takes for granted. Streaming formulations are therefore central. Hallo-Live (Li et al., 2026) generates avatar video in causal blocks that attend to a short horizon of future phonetic cues, OmniForcing (Su et al., 2026) distills a causal student from a bidirectional teacher to reach real-time joint generation, and SoundReactor (Saito et al., 2025) is the only method in Table 10 marked as online, producing audio frame by frame. The constraint compounds with lip synchronization, whose tolerance is roughly a single frame, so this setting demands the tightest alignment under the least favourable sampling budget.

12.4 Personalization

Personalization conditions p_θ on a specific identity, voice, or style, so that generated or edited content matches a target while remaining coherent across modalities. It cuts across the other three applications rather than standing apart from them: the identity preservation that dubbing requires, the speaker consistency that avatars require, and the style control that content creation requires are the same constraint applied at different points. UniAV-Gen (Zhang et al., 2025) addresses it during generation through face-aware modulation that holds

appearance and timbre consistent, while the editing methods of Section 5 address it as a preservation requirement on an existing subject. Personalization is also where the ethical exposure of Section 13 is sharpest, since the capability that makes a legitimate avatar convincing is the capability that makes an impersonation convincing.

13 Open Problems and Future Directions

Scaling joint models. Reaching the best uni-modal quality on both streams without multiplicative data and compute cost is open; pretraining reuse is the main lever.

Long-horizon coherence. Maintaining correspondence over minutes is unsolved: errors accumulate and identity and scene drift—a problem distinct from short-clip synchronization.

Fine-grained cross-modal control. Controlling which source sounds, when, and how loud—per event, not per clip—is the controllability counterpart of raising S at a fine temporal scale.

Physical plausibility. A clip can be synchronized yet physically wrong—sound mismatching the material or force of its visual event—and joint models often fail physical-commonsense benchmarks (Cui et al., 2026).

Under-explored design space. The empty cells of Table 1 mark directions untried rather than failed: no joint method covered here generates either stream as discrete tokens (§4.1), and audio-to-video generation has only isolated dedicated attempts (§7.2, §7.3); each gap has a structural cause and is a reason to study the problem.

Evaluation gaps. Metrics measure per-modality quality well and correspondence poorly; one tracking human judgment of correspondence as reliably as quality metrics do is missing (Zhou et al., 2026).

Ethics, safety, and watermarking. Synchronized speech-and-face generation lowers the barrier to impersonation, making provenance, watermarking, and detection integral; defenses must treat both streams together.

14 Conclusion

We examined generation and editing of video and audio as three problems over one distribution on audio-visual pairs, organized by a five-axis design taxonomy; the trajectory toward unified backbones

reduces the remaining open problems to raising cross-modal alignment while keeping per-stream quality high. The empty cells of our taxonomies are as informative as the occupied ones: token-based joint generation, dedicated audio-to-video methods, and learned narrative and restoration editing are absent or nearly so for structural reasons, each a concrete opening for future work.

Limitations

This work makes several scoping decisions that bound what it can claim. First, we cover only methods in which at least one of video or audio is an output and the other appears in the pipeline as an input or output; single-modality generation (for example, text-to-video without sound or text-to-audio alone), and audio-visual representation learning, retrieval, and understanding without a generative or editing component, are out of scope except as background. Second, we exclude fully closed commercial systems from the taxonomy of Section 4: systems such as Veo 3 (Google DeepMind, 2025) and Sora 2 (OpenAI, 2025) generate synchronized audio-visual content at or beyond the state of the art, but they disclose neither their representations nor their synchronization mechanisms, so placing them on the design axes would amount to guessing; we instead list user-facing capabilities of commercial systems in Appendix A, and mark partially documented systems (for example, Wan 2.5) only on the axes their reports support. The reader should therefore treat the taxonomy as a map of the *documented* literature, and remember that some of the strongest current systems are absent from it by construction. Third, the field is moving quickly: our coverage reflects the literature through mid-2026, several of the systems covered here are described only in preprints or technical reports whose details may change, and empty regions of our taxonomy may fill rapidly. Finally, the five design axes are complementary rather than mutually exclusive, and assigning a method to a category occasionally requires judgment where papers are ambiguous; the per-method tables record our reading, and the cited sources remain authoritative.

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A Other Taxonomies

This appendix collects the per-method taxonomies that support the main text. Table 9 organizes generation and editing methods by task, conditioning signal, and audio target; Table 10 details controls, deployment constraints, and modeling mechanisms for the video-to-audio and Foley family; Table 11 summarizes the dominant modeling mechanism per method; Tables 12 and 13 cover the video-editing methods most relevant to audio-video pipelines; Table 14 lists user-facing capabilities of commercial and foundation systems; and Table 15 records per-method code availability with repository links, each verified individually in July 2026.

Table 9: **Unified taxonomy of video-audio generation and editing methods.** **TASKS** distinguish whether a method generates audio-video from text (T2AV), synthesizes audio for a given video (V2A/Foley), synthesizes video from audio (A2V), generates video and audio jointly (Joint), or supports editing. **CONDITIONING SIGNALS** capture the user/model inputs used to steer generation. **AUDIO TARGETS** indicate what acoustic layers are explicitly modeled. A check mark indicates that the method supports the corresponding capability.

Method	TASKS					CONDITIONING SIGNALS					AUDIO TARGETS				Primary Domain	
	T2AV	V2A/Foley	A2V	Joint	Editing	Text	Video	Audio	Image/Ref.	Spatial Ctrl.	Temporal Ctrl.	SFX/Foley	Speech	Music		Ambience
Visually Indicated Sounds (Owens et al., 2016)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	×	Foley / impacts
Visual to Sound (Zhou et al., 2018)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	×	in-the-wild SFX
Visually Aligned Sound (Chen et al., 2020b)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	×	general V2A
Foley Music (Gan et al., 2020)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	video-to-music
Rhythmic Soundtracks (Gan et al., 2021)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	human movement
Controllable BGM Transformer (Di et al., 2021)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	background music
Dance2Music Q-GAN (Zhu et al., 2022)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	dance music
MM-Diffusion (Ruan et al., 2023)	×	✓	✓	✓	×	×	✓	✓	×	×	✓	✓	×	✓	✓	sounding video
Diff-Foley (Luo et al., 2023)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	×	Foley / SFX
Foley Analogies (Du et al., 2023)	×	✓	×	×	✓	×	✓	✓	×	×	✓	✓	×	×	×	reference-guided Foley
Video BGM Generation (Zhuo et al., 2023)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	background music
Long-Term Rhythmic Soundtracker (Yu et al., 2023)	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×	long rhythmic music
Seeing-and-Hearing (Xing et al., 2024a)	✓	✓	×	✓	×	✓	✓	×	✓	×	✓	✓	×	✓	✓	open-domain AV
V2A-Mapper (Wang et al., 2024a)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	foundation mapper
SonicVisionLM (Xie et al., 2024)	×	✓	×	×	×	✓	✓	×	×	×	✓	✓	×	×	✓	VLM-guided audio
Video-Foley (Lee et al., 2025)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	×	RMS-conditioned Foley
FoleyCrafter (Zhang et al., 2026)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	×	×	✓	text-controlled Foley
Frieren (Wang et al., 2024d)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	efficient V2A
MaskVAT (Pascual et al., 2024)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	masked V2A
STA-V2A (Ren et al., 2024b)	×	✓	×	×	×	✓	✓	×	×	×	✓	✓	×	×	✓	semantic/temporal V2A
VATT (Liu et al., 2024d)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	×	×	✓	caption-mediated V2A
MM-LDM (Sun et al., 2024)	✓	✓	×	✓	×	✓	✓	✓	×	×	✓	✓	×	✓	✓	sounding video
Movie Gen (Polyak et al., 2024)	✓	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	media foundation model
Google V2A (Google DeepMind, 2024)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	✓	✓	✓	video soundtracks
MMAudio (Cheng et al., 2025)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	×	✓	✓	fast V2A / T2A
Mel-QCD (Wang et al., 2025a)	×	✓	×	×	×	✓	✓	×	×	×	✓	✓	×	×	✓	mel-control V2A
VAFlow (Wang et al., 2025b)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	flow matching V2A
Foley-Flow (Mo and Song, 2025)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	masked AV flow
MultiFoley (Chen et al., 2025)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	✓	professional Foley
VinTAGe (Kushwaha and Tian, 2025)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	×	✓	✓	holistic audio
TARO (Ton et al., 2025)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	×	✓	onset-aware V2A
ThinkSound (Liu et al., 2025)	×	✓	×	×	✓	✓	✓	×	×	✓	✓	✓	×	×	✓	interactive audio edit
Hear-Your-Click (Liang et al., 2025)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	✓	click/object V2A
SeIVA (Lee et al., 2026)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	×	selective V2A
MMAudioSep (Takahashi et al., 2026)	×	✓	×	×	✓	✓	✓	✓	×	✓	✓	✓	×	✓	✓	video/text queried separation
CoherentAveEdit (Ishii et al., 2025)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	×	✓	✓	audio follows video edit
AV-Link (Haji-Ali et al., 2025)	×	✓	✓	×	✓	×	✓	✓	×	×	✓	✓	×	✓	✓	bidirectional AV
JavisDiT (Liu et al., 2026a)	✓	×	×	✓	×	✓	×	×	×	×	✓	✓	×	✓	✓	prompt-to-AV
UniAVGen (Zhang et al., 2025)	✓	✓	✓	✓	✓	✓	×	✓	✓	×	✓	✓	×	×	✓	unified AV
RFLAV (Ergasti et al., 2025)	✓	×	×	✓	×	✓	×	×	×	×	✓	✓	×	✓	✓	long/infinite AV
SoundReactor (Saito et al., 2025)	×	✓	×	×	×	×	✓	×	×	×	✓	✓	×	✓	✓	online/game V2A
Sora 2 (OpenAI, 2025)	✓	×	×	✓	✓	✓	✓	✓	×	×	✓	✓	✓	✓	✓	commercial video+audio
Veo 3/3.1 (Google DeepMind, 2025)	×	✓	×	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	commercial video+audio
Adobe Firefly Audio (Adobe, 2025)	×	✓	×	×	✓	✓	✓	✓	×	×	✓	✓	✓	✓	✓	commercial sound design
Seedance 2.0 (ByteDance Seed, 2026)	✓	×	×	✓	✓	✓	✓	✓	×	×	✓	✓	✓	✓	✓	commercial multimodal video
Foley-Omni (Tao et al., 2026)	×	✓	×	×	✓	✓	✓	×	×	×	✓	✓	✓	✓	✓	full video soundtrack
MOVA (SII-OpenMOSS Team, 2026)	✓	×	×	✓	×	✓	×	×	×	×	✓	✓	✓	✓	✓	scalable synchronized AV

Table 10: **Fine-grained taxonomy of video-to-audio, Foley, and audio-following-video-edit methods.** **CONTROLS** summarize how a user or upstream system steers the generated soundtrack. **DEPLOYMENT / EDITING** constraints distinguish long-form, stereo/spatial, online, and source-preserving settings. **MODELING MECHANISMS** summarize the dominant generator or alignment mechanism.

Method	CONTROLS				DEPLOYMENT / EDITING				MODELING MECHANISM						
	Text	Audio Ref.	Click/Mask	Onset/Rhythm	Long-form	Stereo/Spatial	Online	Preserve Src.	LDM	Flow/RF	DiT	AR/Masked	ControlNet	Guidance/Adapter	MLLM/CoT
Diff-Foley (Luo et al., 2023)	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
Foley Analogies (Du et al., 2023)	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
Seeing-and-Hearing (Xing et al., 2024a)	✓	✓	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
V2A-Mapper (Wang et al., 2024a)	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
Video-Foley (Lee et al., 2025)	✓	✓	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗
FoleyCrafter (Zhang et al., 2026)	✓	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗
Frieren (Wang et al., 2024d)	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗
MaskVAT (Pascual et al., 2024)	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗
STA-V2A (Ren et al., 2024b)	✓	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
VATT (Liu et al., 2024d)	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓
Google V2A (Google DeepMind, 2024)	✓	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Movie Gen Audio (Polyak et al., 2024)	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
MMAudio (Cheng et al., 2025)	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗
Mel-QCD (Wang et al., 2025a)	✓	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗
VAFLOW (Wang et al., 2025b)	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗
Foley-Flow (Mo and Song, 2025)	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗
MultiFoley (Chen et al., 2025)	✓	✓	✗	✓	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
ThinkSound (Liu et al., 2025)	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
Hear-Your-Click (Liang et al., 2025)	✓	✗	✓	✓	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
SelVA (Lee et al., 2026)	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
MMAudioSep (Takahashi et al., 2026)	✓	✓	✓	✓	✗	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗
CoherentAVEdit (Ishii et al., 2025)	✓	✓	✗	✓	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗
AV-Link (Haji-Ali et al., 2025)	✓	✓	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗
SoundReactor (Saito et al., 2025)	✗	✗	✗	✓	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗	✗

Table 11: **Modeling-mechanism taxonomy for representative audio-video generation and editing methods.** The table bridges the task-level taxonomies and method sections: it identifies whether a method primarily uses GAN/VQ, autoregression, latent diffusion, flow/rectified-flow, masked modeling, AV alignment losses, joint denoising, frozen-model guidance/adapters, or MLLM/agent-style reasoning.

Method	GAN/VQ	AR/Transformer	LDM	Flow/RF	Masked	AV Align.	Joint Denoise	Frozen/Guidance	MLLM/Agent	Main Mechanism
MM-Diffusion (Ruan et al., 2023)	×	×	×	×	×	×	✓	×	×	joint multi-modal U-Net
Diff-Foley (Luo et al., 2023)	×	×	✓	×	×	×	✓	×	×	CAVP + latent diffusion
Seeing-and-Hearing (Xing et al., 2024a)	×	×	✓	×	×	×	×	✓	×	diffusion latent aligner
V2A-Mapper (Wang et al., 2024a)	×	×	×	×	×	×	×	✓	×	foundation-model mapper
Video-Foley (Lee et al., 2025)	×	×	✓	×	×	×	×	✓	×	RMS two-stage control
FoleyCrafter (Zhang et al., 2026)	×	×	✓	×	×	×	×	✓	×	semantic adapter + temporal controller
Frieren (Wang et al., 2024d)	×	×	×	✓	×	×	✓	×	×	rectified flow matching
MaskVAT (Pascual et al., 2024)	×	✓	×	×	✓	✓	×	×	×	masked generative transformer
STA-V2A (Ren et al., 2024b)	×	×	✓	×	×	✓	×	×	×	local/global visual features
VATT (Liu et al., 2024d)	×	×	✓	×	×	×	×	✓	✓	caption-mediated generation
Movie Gen (Polyak et al., 2024)	×	✓	×	✓	×	×	✓	×	×	scaled media transformers
Google V2A (Google DeepMind, 2024)	×	×	✓	×	×	×	×	✓	×	prompt-conditioned diffusion
MMAudio (Cheng et al., 2025)	×	×	×	✓	×	✓	×	×	×	flow matching + synchronization module
Mel-QCD (Wang et al., 2025a)	×	×	✓	×	×	✓	×	×	×	mel decomposition + ControlNet
VAFLOW (Wang et al., 2025b)	×	×	×	✓	×	×	×	×	×	cross-modal flow matching
Foley-Flow (Mo and Song, 2025)	×	×	×	✓	✓	✓	✓	×	×	masked AV alignment + dynamic flow
MultiFoley (Chen et al., 2025)	×	✓	×	×	×	✓	×	×	×	DiT + multi-conditional training
ThinkSound (Liu et al., 2025)	×	✓	×	×	×	×	×	✓	✓	MLLM CoT + interactive editing
AV-Link (Haji-Ali et al., 2025)	×	×	×	✓	×	×	×	✓	×	frozen flow-model feature links
JavisDiT (Liu et al., 2026a)	×	✓	×	×	×	✓	✓	×	×	joint DiT + spatio-temporal prior
UniAVGen (Zhang et al., 2025)	×	✓	×	×	×	✓	✓	×	×	dual-branch DiT cross-modal interaction
SoundReactor (Saito et al., 2025)	×	✓	×	×	×	✓	×	×	×	causal transformer + diffusion head

Table 12: **Video editing methods most relevant to audio-video generation pipelines: temporal-adaptation, training-modification, and conditioning-branch families.** These methods edit the visual stream; audio-video systems such as CoherentAVEdit can subsequently regenerate or adapt the soundtrack to match the edited result. The columns separate the user control signal, edit target, and implementation strategy.

Year	Method	Family	Instr.	Text	Mask/Box	Point/Traj.	Pose	Image/Style	Motion	Appearance	Inpaint	Tuning-free	Fine-tune	Attention	Latent	Cond. Branch	Canonical
2024	VIA (Gu et al., 2024a)	Temporal adaptation	×	✓	✓	×	×	×	✓	✓	×	×	✓	×	×	×	×
2024	Slicedit (Cohen et al., 2024)	Temporal adaptation	×	✓	×	×	×	×	×	✓	×	✓	×	×	✓	×	×
2024	Factorized Diffusion Distillation (Singer et al., 2024)	Temporal adaptation	×	✓	×	×	×	×	✓	✓	×	×	✓	×	×	×	×
2024	MaskINT (Ma et al., 2024)	Temporal adaptation	✓	✓	✓	×	×	×	×	✓	✓	×	✓	×	×	✓	×
2023	Fairy (Wu et al., 2024a)	Temporal adaptation	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×	×	×
2024	VidToMe (Li et al., 2024b)	Temporal adaptation	×	✓	×	×	×	×	×	✓	×	✓	×	×	✓	×	×
2024	SimDA (Xing et al., 2024b)	Temporal adaptation	×	✓	×	×	×	×	✓	✓	×	×	✓	×	×	✓	×
2023	Text-to-Image Diffusion Video Editing (Zhang et al., 2023b)	Temporal adaptation	×	✓	×	×	×	×	×	✓	×	×	✓	✓	×	×	×
2023	Tune-A-Video (Wu et al., 2023a)	Temporal adaptation	×	✓	×	×	×	×	✓	✓	×	×	✓	×	×	×	×
2026	TrajectoryMover (Chhatre et al., 2026)	Conditioning branch	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	×	×	✓	×
2025	VACE (Jiang et al., 2025)	Conditioning branch	✓	✓	✓	×	×	×	✓	✓	✓	×	✓	×	×	✓	×
2025	VideoPainter (Bian et al., 2025)	Conditioning branch	✓	✓	✓	×	×	×	✓	✓	✓	×	✓	×	×	✓	×
2024	StableV2V (Liu et al., 2024a)	Conditioning branch	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×
2024	EVA (Yang et al., 2024b)	Conditioning branch	✓	✓	✓	×	×	×	×	✓	×	✓	×	×	×	✓	×
2024	Diffutoon (Duan et al., 2024a)	Conditioning branch	×	✓	×	×	×	✓	×	✓	×	×	✓	×	×	✓	×
2024	FlowVid (Liang et al., 2024a)	Conditioning branch	×	✓	×	×	×	×	✓	✓	×	×	✓	×	×	✓	×
2024	AVID (Zhang et al., 2024b)	Conditioning branch	✓	✓	✓	×	×	×	×	×	✓	×	✓	×	×	✓	×
2023	Motion-Conditioned Image Animation (Yan et al., 2023)	Conditioning branch	×	×	×	×	×	✓	✓	×	×	×	✓	×	×	✓	×
2024	LAMP (Wu et al., 2024b)	Conditioning branch	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×
2024	Ground-A-Video (Jeong and Ye, 2024)	Conditioning branch	✓	✓	✓	×	×	×	×	✓	×	✓	×	×	×	✓	×
2023	CCEdit (Feng et al., 2024)	Conditioning branch	✓	✓	✓	×	×	×	×	✓	✓	×	✓	×	×	✓	×
2023	MagicEdit (Liew et al., 2023)	Conditioning branch	✓	✓	✓	×	×	×	×	✓	✓	×	✓	×	×	✓	×
2023	VideoControlNet (Hu and Xu, 2023)	Conditioning branch	×	✓	×	×	×	×	✓	✓	×	×	✓	×	×	✓	×
2023	VideoComposer (Wang et al., 2023b)	Conditioning branch	×	✓	×	×	×	✓	✓	✓	×	×	✓	×	×	✓	×
2023	Structure/Content Guided Video Synthesis (Esser et al., 2023)	Conditioning branch	×	✓	×	×	×	✓	✓	✓	×	×	✓	×	×	✓	×
2024	Movie Gen (Polyak et al., 2024)	Training modification	✓	✓	✓	×	×	✓	✓	✓	✓	×	✓	×	×	✓	×
2024	EffiVED (Zhang et al., 2024a)	Training modification	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×
2024	Customize-A-Video (Ren et al., 2024a)	Training modification	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×
2024	VASE (Peruzzo et al., 2024)	Training modification	✓	✓	✓	×	×	×	×	✓	×	×	✓	×	×	✓	×
2023	Customizing Motion (Materzyńska et al., 2024)	Training modification	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×
2024	SAVE (Song et al., 2024)	Training modification	✓	✓	✓	×	×	×	×	✓	×	×	✓	×	×	✓	×
2024	VMC (Jeong et al., 2024)	Training modification	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×
2024	DreamVideo (Wei et al., 2024)	Training modification	×	✓	×	×	×	✓	✓	×	×	×	✓	×	×	✓	×
2024	Consistent V2V Transfer (Cheng et al., 2024)	Training modification	×	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×
2023	VIDiff (Xing et al., 2023)	Training modification	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×
2024	MotionDirector (Zhao et al., 2024)	Training modification	×	✓	×	×	×	×	✓	×	×	×	✓	×	×	✓	×
2024	InstructVid2Vid (Qin et al., 2024)	Training modification	✓	✓	×	×	×	×	×	✓	×	×	✓	×	×	✓	×
2023	Dreamix (Molad et al., 2023)	Training modification	✓	✓	×	×	×	✓	✓	✓	×	×	✓	×	×	✓	×

Table 13: **Video editing methods most relevant to audio-video generation pipelines: attention/latent/canonical/interactive families.** This continuation covers the attention-injection, motion-feature-injection, latent-manipulation, canonical-representation, point/pose-conditioning, and human(-object)-animation families.

Year	Method	Family	Instr.	Text	Mask/Box	Point/Traj.	Pose	Image/Style	Motion	Appearance	Inpaint	Tuning-free	Fine-tune	Attention	Latent	Cond. Branch	Canonical
2024	VideoGrain (Yang et al., 2025)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	AnyV2V (Ku et al., 2024)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	CoCoCo (Zi et al., 2025)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	Object-Centric Diffusion (Kahatapitiya et al., 2024)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	UniEdit (Bai et al., 2024)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Make-A-Protagonist (Zhao et al., 2023b)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Zero-Shot Video Editing (Wang et al., 2023a)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	FateZero (Qi et al., 2023)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Edit-A-Video (Shin et al., 2023)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Video-P2P (Liu et al., 2024c)	Attention injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	FRESCO (Yang et al., 2024a)	Motion feature injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	FLATTEN (Cong et al., 2024)	Motion feature injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	TokenFlow (Geyer et al., 2024)	Motion feature injection	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	STEM-Inv (Li et al., 2024a)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Video ControlNet (Chu et al., 2023)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Control-A-Video (Chen et al., 2024)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Text2Video-Zero (Khachatryan et al., 2023)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	FRAG (Yoon et al., 2024)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	GenVideo (Harsha et al., 2024)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	MotionClone (Ling et al., 2025)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	RAVE (Kara et al., 2024)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	Space-Time Diffusion Features (Yatim et al., 2024)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	DiffSynth (Duan et al., 2024b)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Rerender-A-Video (Yang et al., 2023a)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	ControlVideo (Zhao et al., 2023a)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Pix2Video (Ceylan et al., 2023)	Latent manipulation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Neural Video Fields Editing (Yang et al., 2023b)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	DiffusionAtlas (Chang et al., 2023)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	StableVideo (Chai et al., 2023)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	CoDeF (Ouyang et al., 2024)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	VidEdit (Couairon et al., 2024)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Layered Video Editing (Kasten et al., 2021)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Neural Video Deflickering (Lei et al., 2023)	Canonical representation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	MotionCtrl (Wang et al., 2024e)	Point/pose conditioning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Drag-A-Video (Teng et al., 2023)	Point/pose conditioning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	DragVideo (Deng et al., 2024)	Point/pose conditioning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	VideoSwap (Gu et al., 2024b)	Point/pose conditioning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	Animate Anyone (Hu et al., 2024)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	MagicAnimate (Xu et al., 2024b)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2023	DreamPose (Karras et al., 2023)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	DisCo (Wang et al., 2024c)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	MagicPose (Chang et al., 2024)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	Fashion-VDM (Karras et al., 2024)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	Animate-X (Tan et al., 2025)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	StableAnimator (Tu et al., 2025)	Human animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2024	AnchorCrafter (Xu et al., 2024c)	Human-object animation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 14: **Commercial and foundation-system capabilities for video generation with audio and audio-for-video editing.** These systems are included because they materially define current user-facing capabilities, even when model details or weights are not fully released.

Year	System	T2AV	V2A	A2V	Joint	Video Edit	Speech/SFX	Music/Amb.	Notes
2024	Movie Gen (Polyak et al., 2024)	✓	✓	✗	✓	✓	✓	✓	research model; video, audio, personalization, editing
2024	Google V2A (Google DeepMind, 2024)	✗	✓	✗	✗	✓	✓	✓	research system; video pixels + optional audio prompt
2025	Veo 3/3.1 (Google DeepMind, 2025)	✓	✓	✗	✓	✓	✓	✓	commercial/product; native audio
2025	Sora 2 (OpenAI, 2025)	✓	✗	✗	✓	✓	✓	✓	commercial/product; synchronized dialogue/SFX
2025	Adobe Firefly Audio (Adobe, 2025)	✗	✓	✗	✗	✓	✓	✓	commercial creative tools; sound effects, soundtrack, speech
2026	Seedance 2.0 (ByteDance Seed, 2026)	✓	✗	✗	✓	✓	✓	✓	commercial/product; text/image/video/audio prompts

Table 15: **Code and artifact availability for the methods we cover** (checked July 2026). ✓: official code released, repository listed; ✗: no functional official release — repositories that exist but hold no code or weights (announcement placeholders, samples-only or dataset-only repositories) count as ✗ and are annotated. Availability changes quickly; each row was checked individually at the date above.

Method	Family	Code	Repository / Weights
MM-Diffusion (Ruan et al., 2023)	Joint gen.	✓	https://github.com/researchmm/MM-Diffusion
CoDi (Tang et al., 2023)	Joint gen.	✓	https://github.com/microsoft/i-Code (i-Code-V3)
Seeing-and-Hearing (Xing et al., 2024a)	Joint gen.	✓	https://github.com/yzxing87/Seeing-and-Hearing (V2A released; other tasks pending)
AV-DiT (Wang et al., 2024b)	Joint gen.	✗	—
MM-LDM (Sun et al., 2024)	Joint gen.	✗	placeholder repository only (no code or weights released)
Movie Gen (Polyak et al., 2024)	Joint gen.	✗	— (benchmark data only)
SVG (Ishii et al., 2024)	Joint gen.	✗	—
MMDiSCo (Hayakawa et al., 2025)	Joint gen.	✓	https://github.com/SonyResearch/MMDiSCo
SyncFlow (Liu et al., 2024b)	Joint gen.	✗	—
JavisDiT (Liu et al., 2026a)	Joint gen.	✓	https://github.com/JavisVerse/JavisDiT
JavisDiT++ (Liu et al., 2026b)	Joint gen.	✓	https://github.com/JavisVerse/JavisDiT (shared repository)
BridgeDiT (Guan et al., 2025)	Joint gen.	✓	https://github.com/guankaisi/BridgeDiT
ALIVE (Guo et al., 2026b)	Joint gen.	✗	placeholder repository only (no code or weights released)
Ovi (Low et al., 2025)	Joint gen.	✓	https://github.com/character-ai/Ovi (inference and weights)
UniAVGen (Zhang et al., 2025)	Joint gen.	✓	https://github.com/MCG-NJU/Sora2-mini
Animate-and-Sound (Wang et al., 2025c)	Joint gen.	✗	—
CCL (Ma et al., 2026)	Joint gen.	✗	—
Hallo-Live (Li et al., 2026)	Joint gen.	✓	https://github.com/fudan-generative-vision/Hallo-Live
UniForm (Zhao et al., 2025)	Joint gen.	✗	—
Wan 2.5 (Alibaba, 2025)	Joint gen.	✗	API-only; no public weights (Wan 2.1/2.2 repositories exclude 2.5)
LTX-2 (HaCohen et al., 2026)	Joint gen.	✓	https://github.com/Lightricks/LTX-2
MOVA (SII-OpenMOSS Team, 2026)	Joint gen.	✓	https://github.com/OpenMOSS/MOVA
Apollo (Wang et al., 2026)	Joint gen.	✗	—
3MDiT (Li et al., 2025)	Joint gen.	✗	—
OmniForcing (Su et al., 2026)	Joint gen.	✓	https://github.com/OmniForcing/OmniForcing
Diff-Foley (Luo et al., 2023)	Cross-modal	✓	https://github.com/luosiallen/Diff-Foley
Foley Analogies (Du et al., 2023)	Cross-modal	✓	https://github.com/XYPB/CondFoleyGen
V2A-Mapper (Wang et al., 2024a)	Cross-modal	✗	samples-only repository (no model code)
Video-Foley (Lee et al., 2025)	Cross-modal	✓	https://github.com/jnwnlee/video-foley
FoleyCrafter (Zhang et al., 2026)	Cross-modal	✓	https://github.com/open-mmlab/FoleyCrafter
Frieren (Wang et al., 2024d)	Cross-modal	✓	https://github.com/cyanbx/Frieren-V2A
MaskVAT (Pascual et al., 2024)	Cross-modal	✗	—
STA-V2A (Ren et al., 2024b)	Cross-modal	✓	https://github.com/PolyPerceiver-Lab/STAV2A
VATT (Liu et al., 2024d)	Cross-modal	✓	https://github.com/DragonLiu1995/video-to-audio-through-text
Google V2A (Google DeepMind, 2024)	Cross-modal	✗	— (closed research system)
MMAudio (Cheng et al., 2025)	Cross-modal	✓	https://github.com/hkchengrex/MMAudio
Mel-QCD (Wang et al., 2025a)	Cross-modal	✓	https://github.com/wjc2830/MelQCD-main (training code pending)
VAFLOW (Wang et al., 2025b)	Cross-modal	✗	—
Foley-Flow (Mo and Song, 2025)	Cross-modal	✗	—
MultiFoley (Chen et al., 2025)	Cross-modal	✗	—
TARO (Ton et al., 2025)	Cross-modal	✓	https://github.com/triton99/TARO
ThinkSound (Liu et al., 2025)	Cross-modal	✓	https://github.com/FunAudioLLM/ThinkSound
Hear-Your-Click (Liang et al., 2025)	Cross-modal	✓	https://github.com/SynapGrid/Hear-Your-Click
SeIVA (Lee et al., 2026)	Cross-modal	✓	https://github.com/jnwnlee/selva (training code pending)
SoundReactor (Saito et al., 2025)	Cross-modal	✗	main model unreleased (official eval toolkit and VAE repositories only)
Foley-Omni (Tao et al., 2026)	Cross-modal	✓	https://github.com/NJU-Speech/Foley-Omni
AV-Link (Haji-Ali et al., 2025)	Cross-modal	✗	repository is a project-page template (no model code)
Lang.-Guided AV Edit (Liang et al., 2024b)	Editing	✗	dataset-only repository (no implementation code)
AvED (Lin et al., 2026)	Editing	✓	https://github.com/GenjiB/AVED
EdiDub (Manela et al., 2025)	Editing	✗	results-only repository (no model code)
Object-AVEEdit (Fu et al., 2025)	Editing	✗	— (project page only)
AV-Edit (Guo et al., 2026a)	Editing	✗	placeholder repository only (no code or weights released)
JUST-DUB-IT (Chen et al., 2026)	Editing	✓	https://github.com/justdubit/just-dub-it (archived; folded into LTX-2 pipeline)
EditYourself (Flynn et al., 2026)	Editing	✗	—